



To Open or Not? Copyright Openness Strategies for Competing Publishers in the Age of Generative Artificial Intelligence

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Abstract. As generative artificial intelligence (AI) reshapes information search paradigms, traffic-monetizing news publishers face a strategic trade-off between protecting copyrights to preserve traditional traffic and opening copyrights to capture AI-driven traffic while risking loss of traditional traffic. We develop a game-theoretic model to examine copyright-opening strategies and welfare implications for two competing publishers when facing an AI engine operating under free or paid business models. Our research reveals that under the free-AI scenario, publishers are less willing to open copyrights to companies with highly accurate AI engines. The underlying mechanism is that high accuracy in the AI engine reduces users' motivation to click publisher links for verifying information, generating traffic gains (*AI expansion effect*) that fall short of the losses from traditional user diversion (*AI cannibalization effect*). Especially when publishers possess large traditional user bases, a severe AI cannibalization effect weakens their private incentives to open copyrights. At this time, competing publishers adopt homogeneous equilibrium strategies, either both open copyrights to the AI company or neither does. To improve the accuracy of AI engines, AI companies may introduce user-paid compensation mechanisms to encourage publishers to open their copyrights. This mechanism can indeed expand the open copyright equilibrium region, which improves consumer surplus and social welfare compared to the closed copyright scenario. Moreover, under the paid AI scenario, compensation for traditional users lost by the non-opening publisher is transferred to the opening publisher (*compensation transfer effect*). This transfer fosters an asymmetric equilibrium in which one publisher opens its copyright while the other does not. Such an equilibrium does not arise under free AI. Our study examines the impact of free versus paid AI engines on the publishing industry, which provides decision-making references for publishers' copyright openness, and calls on policymakers to encourage AI companies to implement targeted traffic-based compensation mechanisms to improve aggregate welfare.

Keywords: generative AI, publisher competition, copyright openness, traffic diversion

1 Introduction

The primary revenue model for traditional news publishers has long depended on direct user traffic to advertising-supported websites. However, with the rapid advancement of generative artificial intelligence (AI) technology, an increasing number of users are relying on AI search engines provided by AI companies—such as ChatGPT by OpenAI and Perplexity AI by Perplexity—to access information. Publishers have keenly observed this trend and are now considering licensing their content copyrights to AI companies to encourage the inclusion of source links in AI-generated answers. Users of AI search engines, motivated by the need for source verification, may click these links, thereby generating incremental traffic and advertising revenue for publishers. By contrast, if publishers refuse to grant content licenses, AI engines may be unable or unwilling to provide links to the original articles. In practice, AI companies have already faced litigation for linking to publishers’ content without authorization. Perplexity AI faced accusations from Condé Nast titles, including Wired and Forbes (Knibbs, 2024), and lawsuits from News Corp outlets, including The Wall Street Journal and New York Post (Dawn & Katie, 2024), for generating answers using news articles without authorization while including publisher links.

Although opening content rights to AI companies enables publishers to capture potential gains from incremental AI-driven traffic, the strategy also embeds the risk of cannibalizing incumbent traditional users. Without publisher authorization, many AI companies’ random and selective scraping of news content frequently causes AI engines to produce “hallucinations”—outputting inaccurate or even fabricated information (Warren, 2025). For instance, Perplexity AI paraphrases two authentic passages from the unauthorized New York Post while additionally fabricating five paragraphs through hallucination (Knibbs, 2024). Such highly inaccurate outputs, typically lacking verifiable source links, substantially raise users’ information verification costs and therefore fail to attract accuracy-oriented traditional users. In contrast, formal publisher authorization enables AI companies to substantially reduce engine inaccuracies through Retrieval-Augmented Generation (RAG) technology and enhance credibility through providing direct links to original sources, thereby repositioning AI engines as more reliable information channels (AI Index Steering Committee, 2025). For example, content authorization granted by News Corp in 2024 allowed OpenAI to access authoritative outlets such as The Wall Street Journal, significantly improving the accuracy of information provided by ChatGPT (News Corp, 2024). At this point, highly reliable and ad-free AI services are capable of inducing accuracy-seeking traditional users to migrate. Gart-

ner (2024) projects that such migration will reduce search traffic to publishers' websites by 25% by 2026. Evidently, copyright-opening decisions confront publishers with a complex trade-off between expanding AI user traffic and the erosion of their traditional user base.

In practice, publishers facing the opportunities and challenges posed by AI engines exhibit sharply divergent strategies regarding copyright openness. Some publishers prioritize potential traffic gains from AI-driven distribution and therefore grant licenses. For example, Axel Springer (2024) entered a global licensing agreement with OpenAI to distribute content via AI channels and capture traffic feedback. In contrast, other publishers, concerned about cannibalizing traditional users, choose to withhold copyrights. Notably, when confronted with OpenAI, The New York Times refused to engage in copyright licensing negotiations and even filed a copyright infringement lawsuit against OpenAI (Stempel, 2023). Such strategic divergence in competitive settings raises a central question: when one publisher opens while another remains closed, although the latter may preserve short-run exclusivity over traditional users, does the loss of access to an emerging AI-based distribution channel risk long-run marginalization? These heterogeneous strategic choices motivate our examination of the underlying drivers of publishers' copyright openness decisions. Furthermore, given the profound influence of AI on information dissemination, clarifying the implications of copyright openness for consumer surplus and social welfare is also essential.

To improve the accuracy of AI models, some AI companies are shifting from purely free services to paid models and introducing user-paid compensation mechanisms that provide direct monetary compensation to publishers who open copyright, thereby mitigating publishers' concerns about traffic cannibalization. For example, Perplexity (2025) launches the "Comet Plus" program, charging users subscription fees while allocating a portion of revenue to licensed publishers in proportion to traffic, aiming to build a sustainable copyright ecosystem. This mechanism theoretically reshapes publishers' incentives for copyright openness. Empirical observations reveal that even after the mechanism's implementation, publishers' strategic responses remain markedly divergent. For example, Wired under Condé Nast swiftly joined the program, hoping to offset losses from traditional user attrition through new revenue streams, whereas The Wall Street Journal and New York Post, under News Corp, have delayed participation, continuing to withhold copyright access. Under competition, the introduction of a user-paid compensation mechanism fundamentally changes publishers' decisions regarding copyright opening in the free-AI scenario. This change also affects the impact of publishers' optimal decisions on consumer surplus and social welfare in the free AI scenario. The aforementioned chain reaction of repercussions is a critical issue that the academic

community needs to research promptly.

In particular, we address three questions: (i) Under the free-AI scenario, when do competing publishers choose to open their copyrights to an AI company? (ii) How does copyright openness toward an AI company affect consumer surplus and social welfare? (iii) When opening copyright to an AI company increases consumer surplus and social welfare but reduces publishers' profits, how does a user-paid compensation mechanism introduced by the AI company reshape the copyright openness game among competing publishers, and can such a mechanism induce copyright opening and achieve mutually beneficial outcomes?

To address these questions, we develop a game-theoretic model in which two competing news publishers provide content services to users with heterogeneous preferences through their websites, monetizing user traffic into advertising revenue. Before determining the number of advertisements (ads) on their websites, each publisher needs to decide whether to open content copyrights to an AI company operating a free AI engine. For publishers, opening copyrights generates incremental traffic from the AI channel but induces diversion of their traditional users to the AI engine, whereas not opening preserves the traditional market while forgoing potential AI-driven traffic gains.

Our research reveals a counterintuitive finding: publishers may prefer to open copyrights to the AI company with a less accurate AI engine. This seemingly altruistic behavior, which favors weaker rather than stronger AI companies, arises from nuanced underlying traffic dynamics. Specifically, an AI engine with high inaccuracy induces stronger user verification incentives, increasing click-throughs to publishers' websites and thereby raising advertising exposure (a stronger *AI expansion effect*), while simultaneously being less effective at diverting traditional users from publishers' sites (a weaker *AI cannibalization effect*). The traffic gains from the former outweigh the losses from the latter, creating a net traffic gain that incentivizes publishers to open copyright to such AI companies. Furthermore, publishers with large traditional user bases also lack the incentive to open copyrights due to the heightened risk of user churn. In a competitive equilibrium, both publishers race to open copyrights to the AI company that generates net traffic gains. This mechanism aligns closely with industry practice. For example, given that AI content generated by ChatGPT is substantially more accurate than that produced by earlier Google Search Generation Experience (SGE), publishers such as The Guardian and The New York Times block OpenAI's GPTBot rather than Google's Googlebot, thereby opening copyrights to Google instead of OpenAI (Milmo, 2023). Our research thus explains why emerging AI companies in the technology catch-up phase are more likely to gain publishers' copyright access than technologically mature AI companies.

Our analysis also shows that publishers opening copyright to the AI company operating a free AI engine always increases consumer surplus. On the one hand, the introduction of the AI engine intensifies competition in traditional markets, compelling publishers to reduce ad impressions and thereby increasing utility for traditional users. On the other hand, the infusion of copyrighted content improves AI engine accuracy, lowering risk costs for AI users. Together, these factors drive growth in consumer surplus. However, aggregate welfare increases only when AI inaccuracy is sufficiently high or sufficiently low: under high inaccuracy, publishers can benefit directly, whereas under low inaccuracy, the consumer surplus gain dominates publishers' profit losses.

To improve the accuracy of AI engines, some AI companies have shifted to a paid model that incentivizes publishers to open copyrights by compensating them for traffic losses caused by AI-driven user diversion. This incentive logic has been validated in practice. For example, although Condé Nast Group sued Perplexity for copyright infringement in 2024, the former later encouraged several of its publications, including The New Yorker, to join Perplexity's Comet Plus program in 2025 to monetize copyright openness (Knibbs, 2024; Perplexity, 2025). Our research indicates that a user-paid compensation mechanism can indeed expand the equilibrium region where publishers open copyrights, and this expanded region also improves consumer surplus and social welfare compared to scenarios where copyrights remain closed. However, when AI engines exhibit extremely high inaccuracy, and publishers possess a weak traditional user base, the mechanism generates a new type of mismatch. Specifically, publishers may be willing to open copyrights, yet this action harms consumer and social welfare. This occurs because compensation revenues induce upward distortion in ad intensity, while high AI inaccuracy amplifies AI users' link-verification behavior, sharply eroding utility for both traditional and AI users. This finding suggests that policymakers should restrict publishers with limited traditional user bases from opening copyrights to low-accuracy AI firms that use a paid model to prevent net welfare losses for consumers. Moreover, compared to free AI engines, this mechanism can transfer compensation corresponding to traditional users lost by the non-open party to the open party (*compensation transfer effect*), which may foster an asymmetric equilibrium with one publisher opening and the other not opening copyrights under high AI inaccuracy.

The remainder of the paper is organized as follows. Section 2 reviews relevant literature and delineates our contributions. Section 3 describes the model setting. Section 4 analyzes the equilibrium in which competing publishers open copyrights to the AI company and the associated welfare implications. Section 5 further examines the changes in the equilibrium results and their social im-

pact after introducing the user-paid compensation mechanism of AI engines. Section 6 summarizes the main findings, discusses managerial implications, and outlines the study’s limitations.

2 Literature Review

This study relates to three major research streams: (1) competition among content publishers; (2) strategic interactions between publishers and content aggregators; and (3) consumer behavior driven by the information quality of AI engines. We review each stream and highlight the gaps between existing studies and our study.

Our study relates to the literature on competition among content publishers. On the market side, prior work examines how competing publishers shape consumer utility through content strategies. For example, Chen & Roels (2024) examine publishers’ competitive strategies via differentiated publishing frequency and find that high-frequency competition driven by digitalization may harm both publisher and consumer welfare. On the advertising side, research focuses on publishers’ optimal responses to advertising competition. Zhang et al. (2024) analyze publishers’ purchases of data insights from data brokers and show that the broker’s exclusive sale of aggregated insights can soften publisher competition while enhancing social welfare. Ding et al. (2025) examine publishers’ advertising strategies in the presence of advertisement blockers and find that competing publishers can achieve a win-win outcome for themselves and consumers by adopting native advertising. Another related stream adopts a two-sided market perspective that integrates the above two sides and studies competition arising from interactions between consumers and advertisers. Athey et al. (2018) find that consumers’ multi-home behavior across competing publishers erodes publisher profits, whereas Amaldoss et al. (2024) reveal that publishers can enhance profits by employing asymmetric pricing strategies to achieve differentiated competition within two-sided markets. Zhang et al. (2024) find that heightened consumer privacy concerns may incentivize publishers to strengthen collaborations with data brokers to boost profits. Our study incorporates emerging AI aggregators into existing research on market-side competition among content publishers. Unlike traditional aggregators such as media platforms, AI engines generate bidirectional user flows by both introducing new users and diverting existing publisher traffic, with these flows endogenously shaped by publishers’ copyright openness decisions. Against this backdrop, our study further examines how competing publishers optimally choose copyright openness strategies when facing free versus paid AI engines and evaluates the implications of these strategies for consumer surplus and

social welfare, thereby offering a new theoretical perspective on publisher copyright strategies in the era of generative AI.

Our study also relates to the literature on strategic interactions between publishers and content aggregators. Prior studies analyze publisher profitability under aggregator-dominated structures. Amaldoss et al. (2024) study optimal content provision when a media platform procures content from publishers with varying competition intensities at a uniform price, showing that a higher advertising share on the platform reduces publisher profits. Johar et al. (2012) demonstrate that, in the exogenous presence of P2P piracy networks, publishers may optimally adopt quality-oriented content strategies to enhance profits. Other studies focus on optimal contract design when publishers and aggregators operate in parallel relationships. Tan & Carrillo (2017) examine strategic choices between agency and wholesale contracts for content publishers and retailers in the digital product context, finding that agency contracts are more likely to achieve a win-win outcome for both publishers and retailers. Furthermore, relevant studies compare the impact of aggregator-dominant and parallel structures on publisher profit. For instance, Cornière & Sarvary (2022) show that allowing publishers to negotiate content prices with platforms yields higher publisher profits than unilateral bundling by platforms. Moreover, some empirical research also documents a two-sided effect when aggregator dominate publisher content. Dellarocas et al. (2016) find that richer information provision by aggregators reduces user click-through to original content, harming publishers, whereas Song & Manchanda (2025) show that platforms benefit from positive spillovers generated by publishers' content for other content on the platform.

The aforementioned literature stream, whether emphasizing publishers being dominated by content aggregators or their parallel relationship, treats the use of publishers' content by aggregators as a given premise, under which to analyze publishers' optimal strategies and profit implications. In contrast, our study endogenizes publishers' strategic positioning by introducing an antecedent decision: whether to open copyrights to a new type of content aggregators—the AI engine. Furthermore, we distinguish between free and paid AI engine business models, revealing how these different models reshape the strategic interactions and social welfare outcomes of competing publishers in their copyright-opening decisions. Thus, this study expands the research perspective of the existing literature that mainly focuses on the given setting in which the publisher's content would be presented by aggregators.

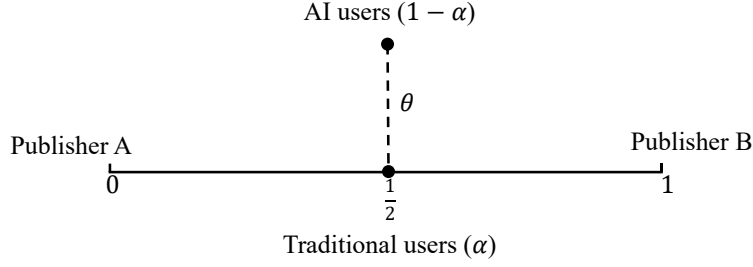
Another related literature stream examines AI applications and their user impacts. Some studies use game-theoretic models to analyze how AI technology affects firms' market strategies and

social welfare. Choi et al. (2024) reveal that improved AI forecasting exerts a nonmonotonic effect on pricing under a ship-then-shop model under a monopoly setting. And Mohammadi et al. (2024) develop a duopoly model to examine regulatory policies for explainable AI and find that mandatory transparency does not necessarily increase social welfare. Other empirical studies document heterogeneous effects of AI-generated content on consumers: generative AI adoption on Q&A platforms significantly enhances readability and response efficiency of user replies (Shan & Qiu, 2025), yet AI's greater penetration in digital content creation may depress consumers' value evaluation (Kyung & Kwon 2022; Rix et al., 2025). These studies primarily treat AI as a firm-level enabling tool and emphasize market strategies of monopolistic or competitive firms driven by AI technology and their impact on social welfare. We extend this stream by re-framing AI engines as content aggregators and studying how competing content publishers—providers of training data—strategically decide whether to open copyrights to AI companies. Such opening behavior entails a nuanced trade-off for publishers: access to AI-driven user traffic revenues versus heightened cannibalization risk for existing users as AI-generated accuracy improves. This perspective not only breaks away from the research paradigm that views AI solely as an enabling tool but also offers new theoretical insights into data source-side strategic choices within the AI ecosystem.

3 Model Setting

We consider two competing news publishers that offer homogeneous content products through their own websites to a geographically dispersed market. Each publisher decides whether to open the content copyright to a company operating a free-to-use AI search engine (hereafter referred to as the “free AI company”). This openness choice directly determines content accessibility for two user types, including traditional and AI users. When neither publisher opens the copyright, traditional users can access full content exclusively through the publisher's website, while AI users can obtain only summaries through the AI channel, without links to the original content products. In contrast, when at least one publisher opens the content copyright, traditional users can access the full content either directly via the publisher's website or indirectly via the AI channel, which provides content summaries and links to the original content products, whereas AI users can both access summaries and their corresponding links through the AI channel. To focus on how channel characteristics and market structure shape incentives for copyright openness, we assume that each user demands at most one content product in this category.

Figure 1. Relative Locations of Users and Publishers



We employ a Hotelling linear city model to characterize the spatial relationship between publishers and users. The market size is normalized to one, among which the traditional users are accounting for α uniformly distributed over the interval $[0, 1]$, with publisher A located at 0 and publisher B at 1, and traditional users' preferences determined by their respective location. The remaining $(1 - \alpha)$ of AI users exhibit no heterogeneous preferences for either publisher's content products, who do not belong to the traditional market but instead concentrate at an independent point directly above the midpoint of the Hotelling line. The perpendicular distance from this point to the Hotelling line segment is $\mathbb{I}_A \mathbb{I}_B \theta$, where $\mathbb{I}_i$, $i \in \{A, B\}$, is an indicator function taking value 1 if publisher i does not open its copyright to the free AI company, and 0 otherwise. The parameter $\theta > 0$ captures AI users' cognitive verification cost, i.e., the effort required to evaluate the reliability of AI-generated summaries when source attribution is unavailable. To obtain a sharp separation between market regimes, we assume that θ is sufficiently large. Hence, when neither publisher opens its copyright ($\mathbb{I}_A = \mathbb{I}_B = 1$), we have $\mathbb{I}_A \mathbb{I}_B \theta = \theta$, which is large enough to make the AI engine an unattractive option for traditional users and they therefore choose only between the two publishers. In contrast, if at least one publisher opens its copyright ($\mathbb{I}_A = 0$ or $\mathbb{I}_B = 0$), then $\mathbb{I}_A \mathbb{I}_B \theta = 0$. This reflects that the openness of copyright enables direct and effective verification of source attribution, thereby eliminating verification costs, which allows AI engines to enter traditional markets as credible external alternatives and potentially replace the role of publishers.

Publishers monetize user traffic by embedding third-party advertisements on their content pages. Let $y_i \geq 0$ denote the number of ads publisher i chooses to embed per page. For example, publishers such as The Guardian and The Financial Times flexibly adjust ad intensity quantities to regulate advertising intensity according to their market strategies (Southern, 2016). To isolate strategic interactions between competing publishers in the consumer market, we adopt an assumption on the advertiser side consistent with Amaldoss & Shin (2025) that advertising supply is abundant relative to user demand, thereby eliminating competition between the two pub-

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lishers in the advertising market. Each ad yields unit revenue (denoted by k) to the publisher while imposing unit disutility (denoted by γ) on users, capturing the mechanism by which ad-induced user annoyance is converted into publisher revenue (e.g., Ren, 2023; Zhang et al., 2024). To isolate strategic interactions triggered by copyright openness that introduces AI channels into the traditional market, we normalize the advertising parameters to $k = \gamma = 1$, which equivalently sets the monetization-to-annoyance ratio $\frac{k}{\gamma}$ to 1. This normalization is without loss of generality, as key insights depend on the relative ratio rather than absolute values, and proportional changes in advertising revenue and user utility can be equivalently captured through adjustments in this ratio.

The user utility is derived jointly by content value and consumption experience. The core value (denoted by v) is obtained when a user successfully gets the target information through any channel, representing the baseline utility from confirming the information itself—for example, whether learning from a full report or a brief notice, confirmation alone generates value v . However, different channels impose varying experiential costs. In the traditional channel, users incur ad annoyance costs of y_i and content mismatch costs, which are determined by their position x on the Hotelling line, measured by the distance x to Publisher A or the distance $1 - x$ to Publisher B. In the AI channel, while AI interfaces are typically ad-free, users face the risk of inaccurate information generated by AI hallucinations. The adverse effect of potential inaccuracy in the AI channel is captured by δ . Given that industry-standard benchmarks (e.g., Vectara, 2026) and extensive international studies (Jaźwińska & Chandrasekar, 2025; Graffius, 2026) have publicly quantified AI hallucination rates δ across AI engines, we assume δ constitutes common knowledge. Notably, δ enters the utility function as an additive disutility term rather than as a reduction in the baseline informational value v . This specification distinguishes two conceptually different components of user payoff: v reflects the benefit from obtaining (and confirming) the target information, whereas δ reflects the perceived risk or discomfort arising from the AI engine’s limited reliability, regardless of whether the realized content turns out to be correct. In particular, when neither publisher opens its copyright and the AI engine cannot provide authoritative attribution, an AI user’s utility is $v - \delta$ —meaning they confirm target information v but bear the full disutility δ associated with the likelihood that the AI information is inaccurate or unverifiable. If only one publisher opens copyrights, AI users may click the source link and obtain utility $v - y_A$ (or $v - y_B$, depending on the open publisher), or refrain from clicking and obtain $v - \lambda\delta$, where $\lambda \in (0, 1)$ denotes the share of copyrighted content not opened to the free AI company—greater copyright openness (smaller λ) lowers the AI engine’s inaccuracy. If both publishers open copyrights, the utility for AI users

clicking the link remains $v - y_A$ or $v - y_B$, while the utility for not clicking further increases to $v - \lambda^2 \delta$, indicating that broader copyright access from more publishers substantially lowers AI inaccuracy. Consequently, publishers' copyright-opening decisions fundamentally reshape AI users' choice sets, shifting them from passively bearing high AI inaccuracy to selecting between low inaccuracy without clicking and certain advertising costs with clicking. Moreover, we assume λ is common knowledge among all market participants, a hypothesis that is consistent with current industry practices of transparent licensing. For example, OpenAI publicly discloses strategic partnerships with leading publishers, including News Corp and Axel Springer, and explicitly specifies the scope of licensed data, encompassing News Corp's complete archival and real-time news feeds (OpenAI, 2024) and Axel Springer's entire portfolio of paid content (OpenAI, 2023). Such public disclosures constitute credible market signals regarding the openness of copyright.

The link-clicking behavior for AI users is driven by their information-verification motive. Such behavior does not aim to seek additional information but reflects distrust of potential AI hallucinations and a desire to verify the accuracy of AI outputs. Naturally, an increase in the AI engine's inaccuracy δ intensifies users' distrust of AI outputs, thereby strengthening their incentives to click links and access original content products for verification. Accordingly, the click-through rate, denoted by β , increases in δ . To model this relationship parsimoniously, we assume β is a linear function of δ , with the proportionality factor determined by the number of publishers that open their copyrights and the extent of such openness. Specifically, we assume $\beta = \lambda \delta$ if a single publisher opens its copyright and $\beta = \lambda^2 \delta$ if two publishers open their copyrights. Here, λ denotes not only the proportion of unopened copyrights but also a trust discount factor: a higher λ implies fewer copyrighted contents opened by publishers, greater AI output inaccuracy, and, for a given δ , stronger AI user distrust and verification-driven clicking. Single openness slightly reduces inaccuracy, lowering the click-through rate from δ to $\lambda \delta$, and double openness further reduces inaccuracy, decreasing the rate to $\lambda^2 \delta$. This formulation implies that as AI content accuracy improves, the AI engine shifts from a user-verified information intermediary to an information endpoint, increasing AI users' propensity to remain within the AI interface.

We assume each publisher chooses between two strategies: opening content copyright to the free AI company (denoted as C) or not opening content copyright (denoted as N). We introduce an index j , where $j \in \{t, s\}$ to represent channel types, with t and s denoting the publisher's traditional channel and AI channel, respectively. We denote the net utility gained by traditional users at location x when accessing publisher i 's content through channel j as U_{ij} , and the number

of such users as d_{ij} . The specific expressions for U_{ij} and the utility function of AI users under different copyright openness configurations are detailed in Table 1, while the specific form of d_{ij} is provided in the Appendix.

Table 1. User Utility in Each Copyright Openness Configuration

Configurations	Traditional users			AI users
	Access A (U_{At})	Access B (U_{Bt})	Convert to AI from A/B (U_{As}/U_{Bs})	—
NN	$v - y_A - x$	$v - y_B - (1 - x)$	—	$v - \delta$
CN	$v - y_A - x$	$v - y_B - (1 - x)$	$v - \lambda\delta$	$\lambda\delta(v - y_A) + (1 - \lambda\delta)(v - \lambda\delta)$
NC	$v - y_A - x$	$v - y_B - (1 - x)$	$v - \lambda\delta$	$\lambda\delta(v - y_B) + (1 - \lambda\delta)(v - \lambda\delta)$
CC	$v - y_A - x$	$v - y_B - (1 - x)$	$v - \lambda^2\delta$	$\lambda^2\delta(v - y_A/y_B) + (1 - \lambda^2\delta)(v - \lambda^2\delta)$

The timing of the game is as follows. In stage 1, competing publishers respectively decide whether to open their content copyrights to the free AI company. In stage 2, two publishers simultaneously determine the number of advertisements y_A and y_B displayed on their content websites in the traditional channel. In stage 3, traditional users make channel selection decisions, choosing which publisher's content to browse through which channel.

4 Equilibrium Analysis

We employ backward induction to solve for the subgame-perfect Nash equilibrium. After publishers make their decisions on content copyright openness (C or N) in stage 1, the game proceeds to four subgames: NN, CN, NC, and CC. Given the symmetry between publishers, CN is equivalent to NC, so NC is omitted. First, in Section 4.1, we analyze the equilibrium ad impressions for publishers in stage 2. Subsequently, in Section 4.2, by comparing three subgames, we obtain the publishers' equilibrium copyright openness strategy in stage 1. Furthermore, we examine the consumer surplus and social welfare of copyright openness strategies under the free-AI scenario in Section 4.3.

4.1 Publishers' Advertising Quantity Decisions in Stage 2

4.1.1 Neither Publisher Opens the Copyright to the AI Company (NN)

We first examine the case in which neither publisher opens the copyright to the free AI company (i.e., $\mathbb{I}_A = \mathbb{I}_B = 0$). In this case, the profit functions for both publishers are $\Pi_A = \alpha y_A d_{At}$ and $\Pi_B = \alpha y_B d_{Bt}$. Solving the game by backward induction yields the equilibrium ad impressions and equilibrium profits in the NN case, as shown in Lemma 1.

Lemma 1. *When neither publisher opens the copyright to the free AI company, the publishers' equilibrium ad impressions and profits are $y_A^{NN} = y_B^{NN} = 1$, $\Pi_A^{NN} = \Pi_B^{NN} = \frac{\alpha}{2}$.*

Analogous to the standard Hotelling model, two symmetric publishers choose identical ad impressions in equilibrium, thereby evenly splitting the traditional user market and earning equal profits. Since traditional users constitute the sole source of profit for both publishers, an increase in the proportion of traditional users, α , directly enhances publisher profitability.

4.1.2 Only One Publisher Opens the Copyright to the AI Company (CN)

We now examine an asymmetric copyright openness decision. Without loss of generality, we assume Publisher A opens the copyright to the free AI company ($\mathbb{I}_A = 0$), while Publisher B does not ($\mathbb{I}_B = 1$), yielding their respective profit functions $\Pi_A = \alpha y_A d_{At} + (1 - \alpha)\lambda\delta y_A$ and $\Pi_B = \alpha y_B d_{Bt}$. In this case, Publisher A benefits from the additional traffic generated by AI users clicking links, while both Publishers A and B face the risk of traditional users switching to AI engines. Similar to the NN case, we can also derive the equilibrium ad impressions and profits under the CN case.

Lemma 2. *When only Publisher A opens the copyright to the free AI company, the publishers' equilibrium ad impressions and profits are*

$$((y_A^{CN}, y_B^{CN}), (\Pi_A^{CN}, \Pi_B^{CN})) = \begin{cases} \left((\lambda\delta, \frac{\lambda\delta}{2}), \left(\lambda^2\delta^2(1 - \alpha), \frac{\lambda^2\delta^2\alpha}{4} \right) \right) & \text{if } \alpha \leq \frac{1}{2} \\ \left((\frac{\lambda\delta}{2\alpha}, \frac{\lambda\delta}{2}), \left(\frac{\lambda^2\delta^2}{4\alpha}, \frac{\lambda^2\delta^2\alpha}{4} \right) \right) & \text{if } \alpha > \frac{1}{2}. \end{cases}$$

Lemma 2 shows that, relative to the equilibrium NN, both publishers choose significantly lower ad impressions under the equilibrium CN because copyright opening introduces an AI engine that directly competes in the traditional market, forcing both publishers to cut ads to curb the loss of traditional users. Here, we partition traditional users into two types: loyal users (who continue using the publisher website) and migrating users (who switch to using the AI engine). Notably, migrating users do not click links on the AI engine to return to publisher websites, as their choice reveals that their expected utility from doing so is strictly dominated by that of using the AI channel. The equilibrium outcome under CN depends on the traditional users' proportion α . When traditional users constitute a small share $\alpha \leq \frac{1}{2}$, the AI engine converts all traditional users into migrating users, leaving no loyal users in the traditional market. Conversely, when $\alpha > \frac{1}{2}$, only a portion of traditional users are converted into migrating users, while loyal users persist in the traditional market. In both scenarios, the ad level of the copyright-opening Publisher A exceeds that of the

non-opening Publisher B ($y_A^{CN} > y_B^{CN}$) because additional traffic from AI users alleviates Publisher A's ad competition pressure in the traditional market. However, AI-induced competition leaves Publisher B unable to share AI traffic benefits while bearing the traditional market erosion, which yields persistently lower profits than Publisher A, highlighting the fragility of the CN equilibrium.

A noteworthy finding is that a stronger traditional user base α may reduce the profits of a publisher that opts for copyright openness. This is illustrated by Publisher A's profit declining as α increases in Lemma 2. This occurs because Publisher A's decision to open copyrights aims to capture traffic revenue from AI users, but simultaneously bears losses from the erosion of its existing traditional users. An increase in α magnifies the latter's losses, causing them to exceed the former's gains, thereby diminishing Publisher A's profits. By contrast, competing Publisher B relies solely on traditional users for profitability that can be directly enhanced by a higher α .

4.1.3 Both Publishers Open Copyrights to the AI Company (CC)

This section analyzes the case in which both publishers open copyrights to the free AI firm (i.e., $\mathbb{I}_A = \mathbb{I}_B = 0$). At this point, the profit functions for Publishers A and B are $\Pi_A = \alpha y_A d_{At} + (\frac{1-\alpha}{2})\lambda^2 \delta y_A$ and $\Pi_B = \alpha y_B d_{Bt} + (\frac{1-\alpha}{2})\lambda^2 \delta y_B$, respectively. Here, the two publishers equally share the incremental traffic generated by AI users' link clicks while jointly bearing the risk of traditional users migrating to the AI engine. Similarly to the NN case, solving the game by backward induction, we can derive the equilibrium ad impressions and profits under the CC case.

Lemma 3. *When both publishers open copyrights to the free AI company, the publishers' equilibrium ad impressions and profits are*

$$((y_A^{CC}, y_B^{CC}), (\Pi_A^{CC}, \Pi_B^{CC})) = \begin{cases} ((\lambda^2 \delta, \lambda^2 \delta), (\frac{\lambda^4 \delta^2 (1-\alpha)}{2}, \frac{\lambda^4 \delta^2 (1-\alpha)}{2})) & \text{if } \alpha \leq \frac{1}{3} \\ ((\frac{\lambda^2 \delta (\alpha+1)}{4\alpha}, \frac{\lambda^2 \delta (\alpha+1)}{4\alpha}), (\frac{\lambda^4 \delta^2 (\alpha+1)^2}{16\alpha}, \frac{\lambda^4 \delta^2 (\alpha+1)^2}{16\alpha})) & \text{if } \alpha > \frac{1}{3}. \end{cases}$$

Lemma 3 demonstrates that when both publishers open copyrights to the free AI company, each publisher's equilibrium ad level falls below that of the copyright-opening publisher under the CN equilibrium, i.e., $y_A^{CC} < y_A^{CN}$. This is because under the CN equilibrium, the exclusive AI user traffic cushions Publisher A's ad competitive pressure in the traditional market and enables it to set more ad impressions. In contrast, under the CC equilibrium, the shared AI traffic weakens Publisher A's ad competitive confidence and reduces its ad level. This lower advertising level also hinders the AI engine's ability to convert all traditional users into migrating users, manifested as a higher threshold for equilibrium structure transition under CC than under CN. Moreover, unlike the CN

equilibrium where a stronger traditional user base α harms only open-access Publisher A, under the CC equilibrium, an increase in α reduces profits for both publishers, i.e., $\frac{\partial \Pi_A^{CC}}{\partial \alpha} = \frac{\partial \Pi_B^{CC}}{\partial \alpha} < 0$. This indicates that after AI competition intervenes, the strengthening of traditional market advantages generally translates into a strategic burden for copyright-opening publishers.

4.2 Publishers' Decisions to Open Copyrights in Stage 1

Thus far, we have analyzed equilibrium ad impressions and profits for publishers in each subgame. This section further compares publishers' profits across subgames to derive the equilibrium outcome of copyright-opening decisions under publisher competition, summarized in Proposition 1. Note that multiple Nash equilibria are coexisting in Proposition 1, and we select the Pareto-dominant equilibrium as the unique equilibrium within this parameter range, that is, the equilibrium in which every publisher earns strictly higher profit than under any alternative equilibrium.

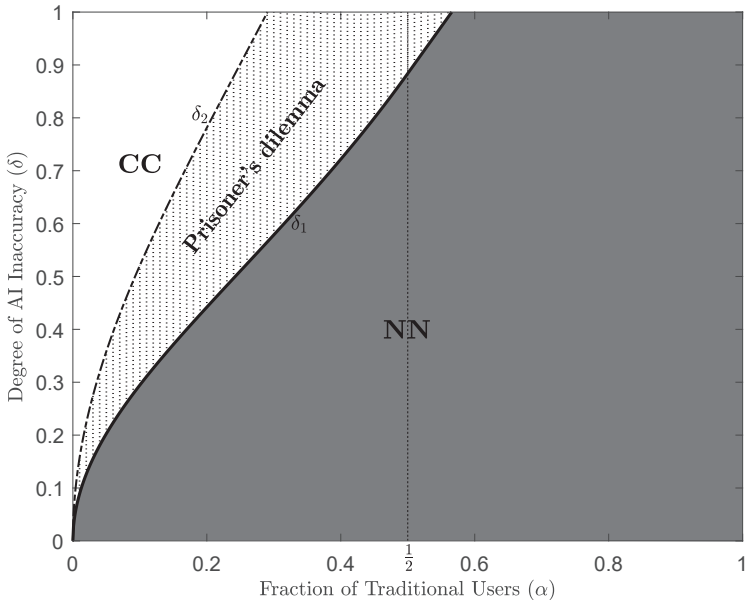
Proposition 1. (Competitive Openness Equilibrium to Free AI). *In equilibrium, neither publisher opens the copyright to the free AI company if the AI engine's inaccuracy is low ($\delta \leq \delta_1$), whereas both publishers open copyright if the AI inaccuracy is high ($\delta > \delta_1$). The threshold is given by $\delta_1 = \frac{\sqrt{2}\sqrt{(1-\alpha)\alpha}}{2(1-\alpha)\lambda}$ if $\alpha \leq \frac{1}{2}$, and $\delta_1 = \frac{\sqrt{2}\alpha}{\lambda}$ otherwise.*

Figure 2 graphically depicts the equilibrium copyright-opening decisions toward the free AI company of publishers under duopoly competition in Proposition 1. Counterintuitively, a higher AI-engine inaccuracy (δ) increases publishers' incentives to open copyrights to the free AI company. Moreover, once such “helping in times of need” behavior emerges, the equilibrium outcome necessarily involves both publishers simultaneously opening their copyrights rather than unilateral action—the asymmetric equilibrium is strictly dominated and thus never arise.

Before analyzing the underlying mechanisms of this competitive equilibrium structure, it is necessary to clarify the key drivers behind publishers' decisions. When a publisher opens its copyright to the free AI company, AI users click through to that publisher's website with probability β . This generates incremental advertising impressions from the AI-user segment, effectively expanding the publisher's market reach from traditional users to AI users. We refer to this positive effect as the *AI expansion effect*. However, the AI engine may also divert traditional users without strong publisher preferences—migrating users positioned adjacent to the Hotelling midpoint—to AI search, reducing visits to publishers' websites, which creates a negative *AI cannibalization effect*. Crucially, once any publisher opens its copyright to the free AI firm, the AI cannibalization effect operates

symmetrically, eroding the traditional user base of both publishers. This occurs because the AI engine becomes a viable, attractive alternative for the same segment of marginal traditional users from both publishers, thereby undermining their localized market power.

Figure 2. Publishers' Equilibrium Copyright Openness Strategies for Free AI ($\lambda = 0.8$)



An increase in AI-engine inaccuracy δ shifts the equilibrium from symmetric non-copyright opening to symmetric copyright opening. When δ is low, high AI accuracy diverts substantial traditional users from the publisher website, while high-quality AI summaries markedly suppress AI users' verification behavior (i.e., the probability of clicking links to copyright-opening publishers). This dual disadvantage of a strong AI cannibalization effect and a weak AI expansion effect reduces publisher incentives to open copyrights, rendering the non-opening equilibrium stable. As δ increases, the AI cannibalization effect weakens and AI expansion effect strengthens, creating incentives for any publisher to move first by opening copyrights to become the sole link source on the AI engine and capture all verification-related gains ($\Pi_A^{CN} > \Pi_A^{NN}$). The rival, however, never tolerates this outcome ($\Pi_B^{CN} < \Pi_B^{CC}$), as non-opening copyright implies both forfeiting incremental AI traffic and exacerbating traditional user losses. To avoid this passive position, the rational rivals always follow suit, driving the equilibrium from asymmetric opening to symmetric opening.

When examining how changes in the share of traditional users, α , affect equilibrium regions, we find that a higher α makes publishers less inclined to open copyrights to the free AI company. As shown in Figure 2, for a given δ , an increase in α shifts the equilibrium from CC to NN. This outcome aligns with expectations, as a stronger traditional user base reduces the incremental traffic

gains from AI users, thereby dampening the AI expansion effect and lowering the attractiveness of copyright openness for publishers. The result also accords with industry practices, where large publishers with substantial local users exhibit stronger resistance to opening copyrights to AI firms compared to small publishers. For example, while dozens of small publishers open copyrights to OpenAI, the established major publisher, The Times, instead pursues litigation against OpenAI and refuses copyright openness (Brown & Jaźwińska, 2025).

Since a higher inaccuracy of the AI engine attracts more AI user traffic for the copyright opening publisher, one might intuitively expect that the profits gained by two publishers opening copyrights to the free AI company with a high-inaccuracy AI engine would always exceed the scenario where neither opens copyright access. However, Corollary 1 examines this conjecture and explicitly refutes this seemingly natural intuition.

Corollary 1. *When $\delta_1 < \delta < \delta_2$, both publishers opening copyrights to the free AI company creates a prisoner's dilemma, where $\delta_2 = \frac{\sqrt{(1-\alpha)\alpha}}{(1-\alpha)\lambda^2}$ if $\alpha \leq \frac{1}{3}$, and $\delta_2 = \frac{2\sqrt{2}\alpha}{(\alpha+1)\lambda^2}$ otherwise.*

Corollary 1 shows that when both publishers open copyrights to a free AI company, the outcome may be inferior to mutual non-opening, leading to a prisoners' dilemma. This arises when AI inaccuracy is intermediate ($\delta_1 < \delta < \delta_2$), resulting from the fact that copyright opening attracts additional AI users but fails to offset losses from traditional users switching to the AI engine. When AI inaccuracy is relatively high ($\delta > \delta_1$), the AI expansion effect strengthens while the AI cannibalization effect weakens, rendering copyright opening a dominant strategy for both publishers. However, once both parties open copyrights, they would split the incremental traffic from AI users' verification, diluting the AI expansion effect. Meanwhile, unlike the scenario where publishers do not open copyrights, they also face the risk of traditional users being eroded by the AI engine. Therefore, as long as the AI inaccuracy is not sufficiently high ($\delta < \delta_2$), the AI cannibalization effect triggered by both publishers opening their copyrights would outweigh the AI expansion effect, resulting in lower profits for two publishers compared to the non-opening scenario.

Moreover, an increase in the proportion of publishers' non-open copyrighted content, λ , lowers the threshold δ_2 . This indicates that the prisoner's dilemma is more likely to arise when publishers open more copyrighted content to AI (i.e., lower λ). Specifically, a decline in λ reduces the payoff of open-copyright publishers through dual channels. On one hand, it increases the utility of migrating users ($v - \lambda^2\delta$), intensifying the AI cannibalization effect. On the other hand, it decreases the probability that AI users click publisher links for verification ($\lambda^2\delta$), weakening the AI expansion

effect. Together, these two channels reinforce the emergence of the prisoner's dilemma.

4.3 The Societal Implications of Copyright Openness

The copyright-opening decisions of two competing publishers to the free AI company affect not only their own profits but also carry significant societal implications. As shown in Section 4.2, a decrease in AI-engine inaccuracy induces an equilibrium shift directly from mutual opening (CC) to mutual non-opening (NN). However, it is important to note that a larger traditional user base (a higher α) also dampens publishers' incentive to open their copyrights, thereby raising the threshold inaccuracy level required to trigger this shift. Since publisher incentives may not align with social objectives, this section examines how this equilibrium transition affects consumer surplus and overall social welfare (defined as the sum of consumer surplus and the two publishers' profits). By comparing the equilibrium NN and CC in terms of consumer surplus and social welfare, we elucidate the social consequences of publishers' copyright-opening strategies and summarize the results in Proposition 2, with the expressions for parameters δ_1^{CS} and δ_2^{CS} provided in the Appendix.

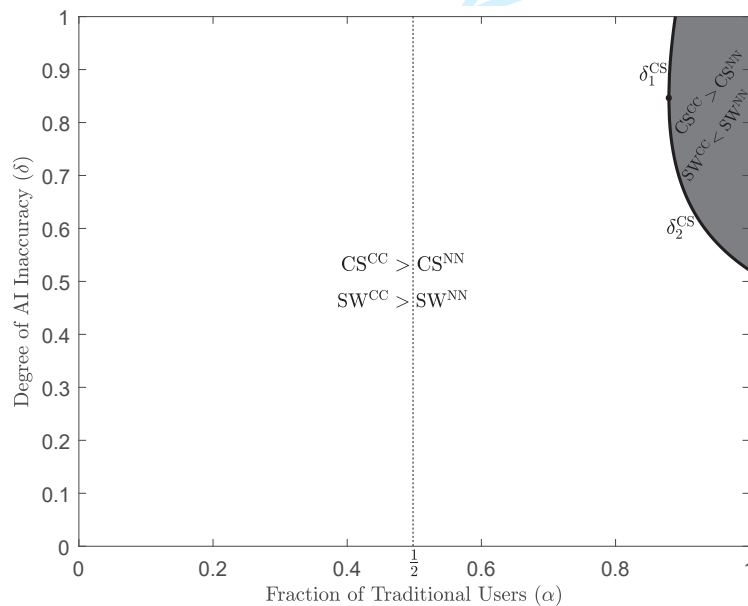
Proposition 2. (Societal Impact of Openness to Free AI). *Compared with the scenario where neither publisher opens the copyright to the free AI company, when both publishers open copyrights to the free AI company, (i) consumer surplus increases; (ii) social welfare increases when δ is high (i.e., $\delta > \delta_1^{SW}$) or low ($\delta < \delta_2^{SW}$) and decreases when δ is moderate (i.e., $\delta_1^{SW} < \delta < \delta_2^{SW}$).*

The results of Proposition 2 are illustrated in Figure 3. A key insight is that publishers' copyright openness to the free AI company always increases consumer surplus, driven by simultaneous improvements for traditional users and AI users. For traditional users, copyright openness introduces a third-party AI engine into the traditional market, intensifying competition and compelling publishers to reduce ad impressions in response to the threat of user migration to the AI engine, thereby lowering the disutility from ad exposure. For AI users, copyright openness reduces AI inaccuracy from δ to $\lambda^2\delta$, mitigating the risk cost associated with distorted AI information. Consequently, the strategy of both publishers opening copyrights raises consumer surplus through two pathways: intensified competition in the traditional market that suppresses advertising interference and improved AI content accuracy that reduces risk costs.

From a social welfare perspective, publishers' copyright openness to the free AI company improves social welfare only when the AI engine's inaccuracy δ is either high or low, stemming from a nuanced trade-off between publisher profit and consumer surplus as δ varies. When δ is high, a

strong AI expansion effect combined with a weak AI cannibalization effect allows publisher profit to remain increased or marginal decline after copyright openness, while the substantial increase in consumer surplus more than offsets the publisher profit change, thereby raising social welfare. When δ is low, although a weak AI expansion effect and strong AI cannibalization effect sharply erode publisher profit, intensified traditional market competition that largely reduces ads and AI accuracy improvement can jointly amplify consumer surplus. These significant gains for consumer surplus exceed publishers' profit losses, yielding a net social welfare increase. By contrast, when δ is moderate, copyright openness substantially harms publisher profit and generates limited consumer surplus gains, so profit losses dominate and social welfare declines (see the gray region of Figure 3). Notably, in the region where δ is relatively low (i.e., $\delta < \delta_1$), although opening copyrights can increase consumer surplus and social welfare, publishers lack incentives to voluntarily open copyrights due to declining profits. The misalignment between publishers' private incentives to open copyrights for profit maximization and societal optimal objectives yields a key managerial implication: market participants that benefit from publishers' copyright openness—such as AI companies that improve AI engine accuracy to attract more users by accessing licensed content—have incentives to design external mechanisms that encourage publishers to open copyrights, potentially alleviating this misalignment.

Figure 3. Social Impact of Copyright Openness for Free AI ($\lambda = 0.8$)



5 Copyright Openness Strategies for Paid AI

Recently, some AI companies have responded to publishers' resistance to copyright openness by designing external incentive schemes to encourage publishers to open copyrights. In June 2024, publications of Condé Nast, including Wired and Forbes, accused Perplexity AI of content copying and issued cease-and-desist letters (Knibbs, 2024). In October of the same year, Dow Jones and the New York Post filed suit against Perplexity, alleging unauthorized reproduction and summarization that erode their traffic and revenue (Dawn & Katie, 2024). In response, Perplexity launches Comet Plus, a revenue-sharing program that charges user subscriptions and allocates a portion of proceeds to publishers that open copyrights, aiming to establish a sustainable compensation mechanism (Perplexity, 2025).

To analyze the incentive effects of the aforementioned user-paid compensation mechanism on publishers' opening copyright decisions, we develop a paid AI model where the company offering a paid AI engine (hereafter, "paid AI company") charges users a subscription fee s and pays a royalty r ($r < s$) to copyright-opening publishers for each additional traditional user redirected to the AI engine, compensating them for lost traffic. We focus on scenarios where the subscription fee s is neither excessively low nor excessively high, satisfying the condition that $\max\{\frac{(r-2\lambda\delta)\alpha+\lambda\delta}{\alpha}, \frac{(r-3\lambda^2\delta)\alpha+\lambda^2\delta}{2\alpha}\} < s < \min\{\frac{(r-3\lambda\delta+2)\alpha+\lambda\delta}{2\alpha}, \frac{(r-3\lambda^2\delta+2)\alpha+\lambda^2\delta}{2\alpha}\}$, to rule out corner cases where copyright opening induces either complete or zero traffic diversion from traditional channels. Under this setting, we re-examine publishers' copyright-opening strategies toward the paid AI company and evaluate the resulting implications for consumer surplus and social welfare. The following presents the main equilibrium results, with complete derivations provided in the appendix. Based on the symmetry of publishers, we compare publishers' profits under the NN, CN, and CC subgames, and arrive at Proposition 3. Consistent with the treatment of equilibrium in Proposition 1, when multiple Nash equilibria coexist, the Pareto-dominant equilibrium is selected as the unique outcome. All parameter expressions involved in each proposition of this section are detailed in the appendix.

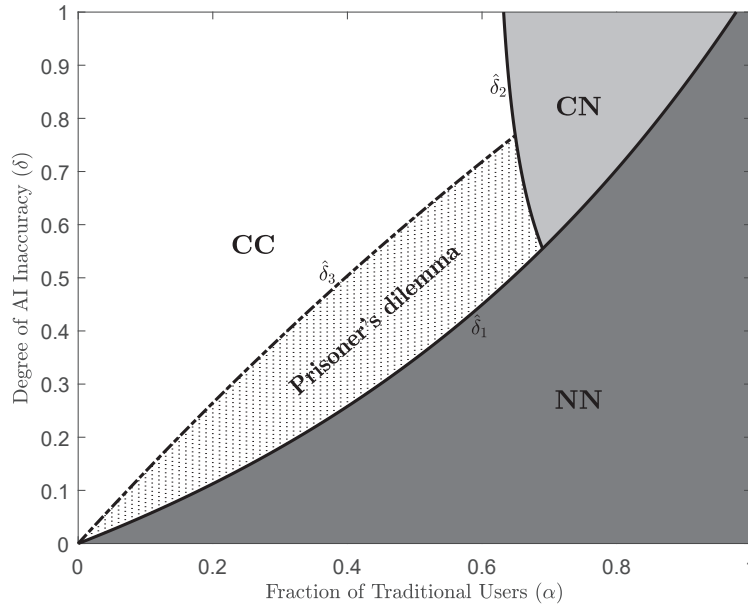
Proposition 3. (Competitive Openness Equilibrium to Paid AI). *In equilibrium, publishers' copyright-opening strategies to the paid AI company are as follows: (i) when $\delta \leq \hat{\delta}_1$, NN is the equilibrium, in which neither publisher opens its copyright to the paid AI company; (ii) when $\hat{\delta}_1 < \delta \leq \hat{\delta}_2$, CC is the equilibrium, in which both publishers open their copyrights to the paid AI company; and (iii) when $\delta > \max\{\hat{\delta}_1, \hat{\delta}_2\}$, CN is the equilibrium, in which only one publisher opens its copyright*

to the paid AI company.

Figure 4 illustrates the equilibrium copyright-opening decisions in Proposition 3 when competing publishers face a paid AI company. As the AI engine's inaccuracy δ increases, the equilibrium shifts from both publishers non-opening (NN) to both opening (CC), and then to only one publisher opening (CN). Unlike the free AI scenario in Proposition 1, where high δ yields only the symmetric opening equilibrium CC, the paid-AI scenario generates the asymmetric equilibrium CN at high δ .

Figure 4. Publishers' Equilibrium Copyright Openness Strategies for Paid AI

($\lambda = 0.8, s = 0.8, r = 0.3$)



In addition to the AI expansion effect and the AI cannibalization effect that publishers should trade off under the free AI scenario, the paid AI setting also introduces a new mechanism. Specifically, publishers who open their copyrights can receive compensation income from the paid AI company for the loss of traditional users, directly offsetting part of the profit decline caused by the AI cannibalization effect. The efficacy of this compensation heavily depends on the rival's choice. When the rival also opens its copyright, compensation is confined to the opening publisher's own lost traditional users. When the rival does not open its copyright, the opening publisher can additionally capture compensation generated by traditional users diverted from its rival, a mechanism referred to as the *compensatory transfer effect*.

To analyze the intrinsic mechanism generating the asymmetric equilibrium under the paid AI scenario, we focus on the region characterized by high AI inaccuracy (i.e., $\delta > \max\{\hat{\delta}_1, \hat{\delta}_2\}$), where

the asymmetric equilibrium emerges, in contrast to the free-AI scenario. Within this region, elevated δ attenuates the AI cannibalization effect while amplifying the AI expansion effect, inducing the focal publisher to open the copyright to the paid AI company. At this point, the focal publisher not only receives compensation for its own lost traditional users but also benefits from the compensation transfer effect for losses of rivals' traditional users. This additional revenue stream enhances the risk tolerance of the focal publisher, thereby prompting an aggressive advertising strategy (i.e., $y_A^{CN} > \max\{y_B^{CN}, y_A^{CC}\}$). Despite accelerated attrition of traditional users due to high ad impressions, the focal publisher can still achieve substantial profits by leveraging compensatory gains derived from eroding traditional users. Meanwhile, the rival faces a strategic trade-off. If following suit with open copyright, the resulting symmetric competition not only eliminates the compensation transfer effect but also triggers intense ad competition in the traditional market, severely damaging profits. Therefore, the rival's optimal response is to maintain copyright closure that though forgoing gains form the AI expansion effect while exploiting the focal publisher's aggressive ad strategy, which weakens its traditional-market appeal and allows the rival to set a relatively moderate ad level to attract substantial ad-sensitive traditional users (i.e., $d_{Bt}^{CN} > \max\{d_{At}^{CN}, d_{Bt}^{CC}\}$) and capture greater ad revenue. Thus, the focal publisher deepens AI side revenue through the compensation transfer and AI expansion effects, whereas the rival concentrates on the traditional market that is being "actively ceded" by the focal publisher to avoid direct competition and sustain ad profits, yielding a stable asymmetric equilibrium.

More importantly, we find that compared to the free AI scenario, the user-paid compensation mechanism under paid AI can effectively incentivize competing publishers to open copyrights ($\hat{\delta}_1 < \delta_1$; see Figures 4 and 2), thereby alleviating the inefficiency under free AI whereby the copyright openness is most favorable to consumer surplus yet publishers refuse to open copyrights. This incentive for copyright openness stems from the fact that publishers who open their copyrights not only mitigate profit losses caused by the AI cannibalization effect through compensatory income, but also benefit from subscription fees generated by AI-induced new users, which attenuate the intensity of the AI cannibalization effect. These dual benefits jointly strengthen publishers' incentive to open their copyrights. Meanwhile, the existence of the compensatory transfer effect can also sustain a stable asymmetric equilibrium with a single open publisher (as discussed above). Furthermore, we also find that a prisoner's dilemma persists under the paid AI scenario (see the hatched region in Figure 4).

Through the preceding analysis, one might conjecture that the user-paid compensation mecha-

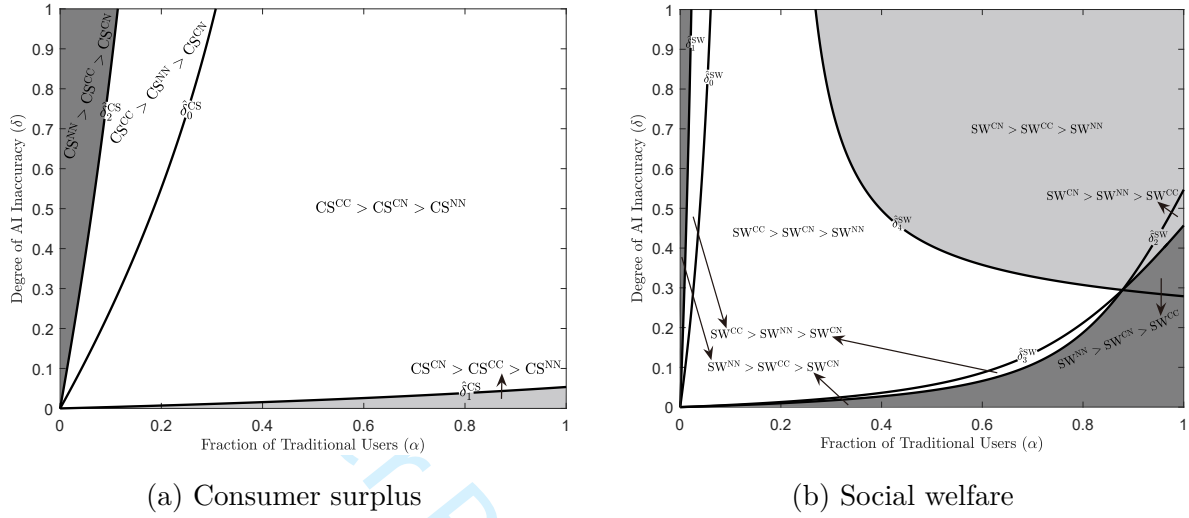
nism introduced by the AI company would affect consumers more subtly by reshaping publishers' advertising incentives. We therefore examine how publishers' copyright-opening decisions under the paid AI scenario influence consumer surplus, with the results formally summarized in Proposition 4.

Proposition 4. (Consumer Surplus under Paid AI). *The impact of competing publishers' decisions to open copyrights to a paid AI company on consumer surplus is as follows: (i) when $\delta \leq \hat{\delta}_1^{\text{CS}}$, the CN mode is most favorable to consumer surplus ($\text{CS}^{\text{CN}} > \text{CS}^{\text{CC}} > \text{CS}^{\text{NN}}$); (ii) when $\hat{\delta}_1^{\text{CS}} < \delta \leq \hat{\delta}_2^{\text{CS}}$, the CC mode is most favorable to consumer surplus ($\text{CS}^{\text{CC}} > \text{CS}^{\text{CN}} > \text{CS}^{\text{NN}}$ or $\text{CS}^{\text{CC}} > \text{CS}^{\text{NN}} > \text{CS}^{\text{CN}}$); (iii) when $\delta > \hat{\delta}_2^{\text{CS}}$, the NN mode is most favorable to consumer surplus ($\text{CS}^{\text{NN}} > \text{CS}^{\text{CC}} > \text{CS}^{\text{CN}}$).*

Unlike the free AI scenario, where open copyright always benefits consumers, in the paid AI scenario, the mode most favorable to consumers shifts sequentially from asymmetric openness (CN) to symmetric openness (CC), and finally to symmetric non-openness (NN) as the AI engine's inaccuracy δ increases, whose dynamic process is shown in Figure 5(a). The core mechanism behind this divergence lies in the introduction of the user-paid compensation mechanism within the paid AI scenario, comprising subscription fees s and compensation income r : the former establishes a new threshold for users accessing information via the AI engine, while the latter distorts publishers' advertising incentives. When δ is low, the CN mode is most favorable to consumer surplus, primarily because the non-opening publisher adopts a low ad strategy that substantially expands and benefits its loyal users. At this point, high AI accuracy implies that migrating users obtain utility from AI use that far exceeds the subscription fee. Meanwhile, copyright openness intensifies AI-induced competition in the traditional market, inducing both publishers to reduce ads to retain traditional users ($y_B^{\text{CN}} < y_A^{\text{CC}} < y_A^{\text{CN}} < y_A^{\text{NN}}$), thereby mitigating ads disutility for loyal users. Moreover, copyright openness also allows AI users to verify content accuracy while encountering fewer ads and achieving greater utility. Although any openness mode is more favorable than NN for all three user types, CN yields higher consumer surplus than CC at low δ . This occurs because in the CN mode, the non-opening publisher sets exceptionally low ads ($y_B^{\text{CN}} < y_A^{\text{CC}}$) to expand loyal user base and markedly increase loyal-user utility, while the opening publisher's higher ads ($y_A^{\text{CN}} > y_A^{\text{CC}}$) reduce its loyal users but with limited utility loss. The resulting net effect is a substantial increase in aggregate loyal-user surplus. For AI users, redirection occurs only to the high-ads opening publisher, yet low δ implies a small jump probability ($\lambda\delta$) and thus negligible expected loss. For migrating users, although AI inaccuracy is slightly higher under CN than CC, a low δ keeps the incremental risk loss ($\delta\lambda(1 - \lambda)$) small. Consequently, the sizable gains to loyal users under CN

outweigh the limited losses to the other two user types, yielding $CS^{CN} > CS^{CC} > CS^{NN}$.

Figure 5. Social Impact of Copyright Openness for Paid AI ($\lambda = 0.8$, $s = 0.8$, $r = 0.3$)



As δ increases to a moderate level, the CC mode is most favorable to consumer surplus because compensation income and AI-user traffic sharing jointly curb the ad distortion under the CN mode. At this time, moderate AI accuracy yields modest net gains for migrating users after subscription fees. Higher AI inaccuracy increases AI users' verification propensity, thereby raising their exposure to publishers' advertising. Under the CN mode, the opening publisher internalizes all link-click traffic from AI users (share $1 - \alpha$) to inflate ads, and combined with the compensation transfer effect, allowing ads to beyond that under the NN mode level ($y_A^{CN} > y_A^{NN}$), severely harming the utility of both loyal and AI users. By contrast, under the CC mode, publishers share AI traffic and compensation income, inducing more restrained ads ($y_A^{CC} < y_A^{NN}$) and benefiting both user types. Meanwhile, dual-source content generation in the CC mode significantly enhances migrating users' risk-mitigation gains by $\delta\lambda(1 - \lambda)$. Consequently, at moderate δ , the CC mode becomes the most favorable mode for consumer surplus, yielding $CS^{CC} > CS^{CN} > CS^{NN}$ or $CS^{CC} > CS^{NN} > CS^{CN}$.

As δ further increases to high levels, the NN mode becomes the consumer-surplus-maximizing mode, under which the upward distortion of ads induced by user-paid compensation schemes for copyright openness is effectively eliminated. When publishers open copyrights, high AI inaccuracy reduces the net surplus of migrating users after subscription payments are made. Simultaneously, high AI inaccuracy (at $\lambda\delta$ under CN mode and $\lambda^2\delta$ under CC mode) substantially increases the AI users' click-through rate, compelling them to extensively visit open-copyright publisher pages and endure high-intensity ad exposure. Regardless of CN or CC mode, compensation fees under high δ distort publishers' advertising incentives, causing open-copyright publishers' ad impressions

to exceed NN mode levels (i.e., $y_A^{CN} > y_A^{CC} > y_A^{NN}$). This ad inflation inevitably amplifies loyal users' disutility, while AI users face heavy ad interference due to a higher probability of verification behavior, severely impairing the utility for both user types. The meager gains from migrating users are far from sufficient to offset these losses. Therefore, any form of copyright openness mode ceases to benefit consumers at high δ , and the NN mode thus becomes the optimal refuge for consumer surplus, having $CS^{NN} > CS^{CC} > CS^{CN}$.

Hence, when AI inaccuracy lies at an intermediate level (i.e., $\hat{\delta}_1 < \delta < \hat{\delta}_2^{CS}$), copyright openness is most favorable to consumer surplus, and meanwhile, there happens to be at least one publisher optimally chooses to open its copyright (see Proposition 3). This result indicates that a paid AI mode can partially reconcile the conflict between publishers' private incentives and consumer surplus of the free AI mode, thereby rendering consumer-surplus-enhancing copyright openness implementable. Several real-world studies validate this mechanism's effectiveness. After accusing Perplexity of copyright infringement and issuing cease-and-desist letters in 2024, Condé Nast subsequently allowed flagship publications such as The New Yorker, Wired, and GQ to join Perplexity's Comet Plus partner program in 2025 (Knibbs, 2024; Perplexity, 2025), under which copyright openness offers high-accuracy summaries for paid AI users while compensating publishers for lost traffic, driving publishers to open copyrights and increasing consumer surplus as compared to non-opening scenarios. Notably, when AI inaccuracy is high (i.e., $\delta > \hat{\delta}_2^{CS}$), copyright opening by both publishers instead harms consumer surplus. This finding yields a crucial managerial implication for policymakers: when competing publishers share weak traditional user bases, opening copyright to low-accuracy paid AI firms should be restricted, as such openness induces excessive advertising expansion by publishers, thereby substantially undermining consumer welfare.

Proposition 4 reveals a nonmonotonic effect of copyright openness on consumer surplus, implying that its effect on aggregate social welfare may be more intricate. This observation raises a critical question: when publishers' private incentives diverge from the mode that most favorable to consumer surplus, can their optimal copyright-opening strategy remain aligned with social welfare objectives? Proposition 5 addresses this question by characterizing the optimal mode for social welfare across different parameter regions and assessing the effectiveness of a user-paid compensation mechanism in reconciling publishers' private incentives with social welfare goals.

Proposition 5. (Social Welfare under Paid AI). *The impact of competing publishers' decisions to open copyrights to a paid AI company on social welfare is as follows: (i) When $\delta > \hat{\delta}_1^{SW}$ or $\delta \leq \min\{\hat{\delta}_2^{SW}, \hat{\delta}_3^{SW}\}$, the NN mode is most favorable to social welfare ($SW^{NN} > SW^{CC} > SW^{CN}$)*

or $SW^{NN} > SW^{CN} > SW^{CC}$); (ii) When $\delta > \max\{\hat{\delta}_3^{SW}, \hat{\delta}_4^{SW}\}$, the CN mode is most favorable to social welfare ($SW^{CN} > SW^{CC} > SW^{NN}$ or $SW^{CN} > SW^{NN} > SW^{CC}$); (iii) When $\hat{\delta}_2^{SW} < \delta \leq \min\{\hat{\delta}_1^{SW}, \hat{\delta}_4^{SW}\}$, the CC mode is most favorable to social welfare ($SW^{CC} > SW^{NN} > SW^{CN}$ or $SW^{CC} > SW^{CN} > SW^{NN}$).

Under the paid AI scenario, the mode most beneficial to social welfare exhibits a complex switching as the AI inaccuracy δ increases, transitioning sequentially from NN to CC to CN and back to NN, as illustrated in Figure 5(b). Specifically, when δ is extremely low, the NN mode is most favorable to social welfare. Although copyright opening raises consumer surplus, weak AI expansion and strong AI cannibalization effects substantially erode both publishers' profits, with profit losses exceeding consumer surplus gains, yielding $SW^{NN} > SW^{CC} > SW^{CN}$ or $SW^{NN} > SW^{CN} > SW^{CC}$. As δ increases slightly, the CC mode becomes optimal. At this point, copyright opening reduces ad nuisance and improves AI content accuracy, sharply increasing consumer surplus, while publishers' profits rise through the compensation transfer and AI expansion effects, generating joint gains in consumer surplus and publisher profits, having $SW^{CC} > SW^{NN} > SW^{CN}$ or $SW^{CC} > SW^{CN} > SW^{NN}$. As δ increases further, the CN mode becomes socially optimal. Within this range, the substantially elevated δ amplifies the AI expansion effect and mitigates the AI cannibalization effect, enabling the copyright-opening publisher to deepen AI markets via the compensation transfer and AI expansion effects while the non-opening publisher focuses on the high-ad-revenue traditional market. This division of labor significantly boosts publishers' overall profits, exceeding the consumer surplus losses caused by high ad impressions and highly inaccurate AI content, implying $SW^{CN} > SW^{CC} > SW^{NN}$ or $SW^{CN} > SW^{NN} > SW^{CC}$. Finally, when δ is extremely high, the NN mode regains the optimum. The AI engine's high inaccuracy makes non-open copyright most beneficial for consumer surplus, while publishers suffer profit losses from missing out on the strong AI expansion effect. The former's gains exceed the latter's losses, yielding $SW^{NN} > SW^{CC} > SW^{CN}$.

The foregoing analysis shows that, under paid AI, the alignment between publishers' copyright decisions and social welfare hinges on AI inaccuracy (δ). When AI inaccuracy is high (i.e., $\delta > \hat{\delta}_1^{SW}$), although AI companies offer compensation to successfully encourage publishers to open copyrights, publishers may distort ad volume upward to offset the erosion of traditional traffic by AI engines, sharply reducing consumer surplus. Thus, in this region, withholding copyright is most favorable to both consumer surplus and social welfare. Accordingly, for technologically immature AI models, policymakers should be alert to advertising inflation induced by licensing agreements and, when

necessary, impose caps on ad intensity to curb publishers' excessive extraction of residual traffic value. When AI inaccuracy is moderate (i.e., $\min\{\hat{\delta}_2^{\text{SW}}, \hat{\delta}_3^{\text{SW}}\} < \delta \leq \hat{\delta}_1^{\text{SW}}$), publishers voluntarily open copyright, and this choice is jointly most beneficial to consumer surplus and social welfare, underscoring the effectiveness of user-based compensation under paid AI. When inaccuracy is low (i.e., $\delta \leq \min\{\hat{\delta}_2^{\text{SW}}, \hat{\delta}_3^{\text{SW}}\}$), publishers' decision to withhold copyrights, though being most favorable to social welfare, harms consumers' interests. In this case, conventional licensing arrangements may fail, and consumer-oriented policymakers may draw on Australia's News Media Bargaining Code (Australian Competition and Consumer Commission, 2021) by adopting mandatory bargaining to ensure that AI-driven efficiency gains are adequately returned to content publishers, thereby further incentivizing copyright opening.

6 Conclusion

As generative AI reshapes the landscape of information distribution, traditional news publishers face an unprecedented strategic choice between cooperation and confrontation. Our study develops a game-theoretic model to examine whether and when competing news publishers should open content copyrights to AI companies amid the rapid diffusion of generative AI search engines. We internalize copyright openness as a strategic choice for competing publishers and examine how it alters the competitive intensity among publishers through induced changes in advertising allocation and user traffic distribution. We also evaluate the welfare implications of competing publishers' strategic behavior in opening copyrights to AI companies.

Our study reveals a counterintuitive finding that publishers are more willing to open their copyrights to AI companies with low-accuracy AI engines. The core mechanism is that high AI inaccuracy increases AI users' verification-driven propensity to click publisher links, transforming the AI interface from an informational endpoint into a potential traffic intermediary. When such click-induced AI expansion effects dominate AI-driven cannibalization of traditional demand, copyright openness becomes a rational strategy for publishers. In practice, publishers such as The Guardian and The New York Times have shown greater willingness to open copyrights to Google SGE, which exhibits lower accuracy, while adopting a more cautious stance toward the more accurate ChatGPT (Milmo, 2023), which aligns precisely with the mechanism we uncovered.

Our welfare analysis reveals a nuanced divergence between publishers' private incentives and social welfare. Although opening copyrights to the free AI company can simultaneously increase

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consumer surplus and overall social welfare when traditional user bases are large, the severe profit erosion caused by the AI cannibalization effect often leaves publishers lacking private incentives to open copyrights. This incentive gap constrains free AI companies’ access to copyrighted content needed to improve training data and refine AI technologies, providing an economic rationale for user-paid compensation mechanisms that offset publishers’ losses and induce copyright openness. For instance, Perplexity launched the paid partner program “Comet Plus” in 2025, charging users subscription fees and compensating opening copyright publishers to mitigate their user traffic diversion losses and induce copyright openness (Perplexity, 2025).

We also find that AI companies’ user-paid compensation mechanisms effectively induce publishers to open their copyrights, thereby expanding the open copyright equilibrium region, which can improve consumer surplus and social welfare compared to non-opening scenarios. However, when AI engines exhibit high inaccuracy and publishers possess a weak traditional user base, although this compensation mechanism still induces copyright opening, consumer surplus and social welfare instead decline. This finding provides a critical regulatory implication: policymakers should restrict publishers with limited traditional user bases from opening copyrights to AI firms adopting paid models but exhibiting low accuracy, thereby preventing welfare losses to consumers. Moreover, this mechanism also fosters a stable asymmetric equilibrium that reshapes the equilibrium structure in the free-AI scenario. The underlying reason is that compensation income attenuates competitive intensity for copyright-opening publishers in traditional ad markets, rendering its rival willing to hold onto moderate ad returns in the traditional market. This theoretical deduction aligns closely with observed polarization in the news industry: Wired under Condé Nast rapidly joins the Comet Plus program by Perplexity (2025), whereas The Wall Street Journal under News Corp declines participation. Meanwhile, this finding carries clear policy implications that regulatory frameworks should move beyond absolute copyright protection toward incentivizing AI firms to deliver targeted compensation for copyright-opening publishers, thereby fostering the co-evolution of generative AI and content production.

Our study has several limitations that warrant future research. First, we focus on competition between two symmetric publishers. Subsequent work can incorporate publisher heterogeneity—such as differences in content quality or brand influence—and examine how such asymmetries reshape the equilibrium of their copyright-opening strategies toward AI companies. Second, our model does not treat the AI company as a strategic actor and takes AI-engine inaccuracy as exogenous. Further research can construct a richer game-theoretic framework that endogenizes the

AI company's accuracy-enhancing investment in the face of competing publishers and analyzes how such investments feed back into publishers' copyright-opening decisions and social welfare. Finally, future research can further extend the market structure to allow multi-homing by publishers or consumers across AI engines with heterogeneous technological capabilities, thereby deepening the understanding of the sustainable generative-AI ecosystem.

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E-Companion

Proof of Lemma 1.

We derive the equilibrium ad impressions and profits in the NN case, where neither publisher opens copyrights to the free AI company. A traditional user located at x obtains utility $U_{At} = v - y_A - x$ from publisher A and $U_{Bt} = v - y_B - 1 + x$ from publisher B. Setting $U_{At} = U_{Bt}$ yields the indifferent consumer point $x^{NN} = \frac{y_B}{2} + \frac{1}{2} - \frac{y_A}{2}$. Accordingly, publishers' user demands are $d_{At}^{NN} = \frac{y_B}{2} + \frac{1}{2} - \frac{y_A}{2}$ and $d_{Bt}^{NN} = 1 - (\frac{y_B}{2} + \frac{1}{2} - \frac{y_A}{2})$. Substituting d_{At}^{NN} and d_{Bt}^{NN} into publishers' profit functions gives $\Pi_A = y_A \alpha (\frac{y_B}{2} + \frac{1}{2} - \frac{y_A}{2})$ and $\Pi_B = y_B \alpha (\frac{1}{2} - \frac{y_B}{2} + \frac{y_A}{2})$. Since $\frac{\partial^2 \Pi_A}{\partial y_A^2} = \frac{\partial^2 \Pi_B}{\partial y_B^2} = -\alpha < 0$, each profit function is concave in its own ad impression. The first-order conditions $\frac{\partial \Pi_A}{\partial y_A} = -\frac{\alpha(-y_B-1+2y_A)}{2} = 0$ and $\frac{\partial \Pi_B}{\partial y_B} = \frac{\alpha(1-2y_B+y_A)}{2} = 0$ yield $y_A^{NN} = y_B^{NN} = 1$. Given the equilibrium ad impressions, we can obtain that the equilibrium profit of each publisher is $\Pi_A^{NN} = \Pi_B^{NN} = \frac{\alpha}{2}$.

Proof of Lemma 2.

Similar to the NN case, we derive the equilibrium ad impressions and profits in the CN case. Publisher A opens its copyright to the free AI company, whereas Publisher B does not. A traditional user located at x obtains utility $U_{At} = v - y_A - x$ from Publisher A, $U_{Bt} = v - y_B - 1 + x$ from Publisher B, or $U_{As}/U_{Bs} = v - \lambda\delta$ by switching to the AI engine via either publisher. For traditional users, equating $U_{At} = U_{As}$ yields the indifferent point $x_A^{CN} = \lambda\delta - y_A$, and equating $U_{Bt} = U_{Bs}$ yields $x_B^{CN} = -\lambda\delta + y_B + 1$. Thus, traditional demands are $d_{At}^{CN} = \lambda\delta - y_A$ and $d_{Bt}^{CN} = \lambda\delta - y_B$. Substituting into the profit functions gives $\Pi_A = -\alpha y_A^2 + \lambda\delta y_A$ and $\Pi_B = y_B \alpha (\lambda\delta - y_B)$. Since $\frac{\partial^2 \Pi_A}{\partial y_A^2} = \frac{\partial^2 \Pi_B}{\partial y_B^2} = -2\alpha < 0$, each profit function is concave in its own ad impression. The first-order conditions $\frac{\partial \Pi_A}{\partial y_A} = -2y_A \alpha + \lambda\delta = 0$ and $\frac{\partial \Pi_B}{\partial y_B} = \alpha(\lambda\delta - 2y_B) = 0$ yield $y_A^{CN} = \lambda\delta$, $y_B^{CN} = \frac{\lambda\delta}{2}$, $\Pi_A^{CN} = \frac{\delta^2 \lambda^2}{4\alpha}$, and $\Pi_B^{CN} = \frac{\lambda^2 \delta^2 \alpha}{4}$. Substituting these ad impressions into the demand and profit expressions shows that, except for $d_{At}^{CN} > 0$ requiring $\alpha > \frac{1}{2}$, all other quantities are unconditionally positive.

Since the above equilibrium requires $\alpha > \frac{1}{2}$, we consider this case when $\alpha < \frac{1}{2}$, in which the AI engine attracts all of Publisher A's traditional users. Publisher A then forgoes the traditional market and gets the profit $\Pi_A = \lambda\delta y_A (1 - \alpha)$. Maximizing this expression implies $y_A^{CN} = \lambda\delta$, meaning that publisher A can maximize the capture of consumer surplus from AI users clicking

on its link from the AI engine. So the resulting optimal profit is $\Pi_A^{CN} = \lambda^2 \delta^2 (1 - \alpha)$ for $\alpha < \frac{1}{2}$. Owing to AI-induced market fragmentation that locally monopolizes each publisher's traditional users, Publisher B's equilibrium ad impression and profit remain the same as in the case $\alpha > \frac{1}{2}$.

Proof of Lemma 3.

In the CC case, both publishers open copyrights to the free AI company. A traditional user located at x obtains utility $U_{At} = v - y_A - x$ from publisher A, $U_{Bt} = v - y_B - 1 + x$ from publisher B, or $U_{As}/U_{Bs} = v - \lambda^2 \delta$ by switching to the AI engine via either publisher. Equating $U_{At} = U_{As}$ yields the indifferent point $x_A^{CC} = \lambda^2 \delta - y_A$, and equating $U_{Bt} = U_{Bs}$ yields $x_B^{CC} = -\lambda^2 \delta + y_B + 1$. Substituting these into the profit functions gives $\Pi_A = \frac{y_A(-2\alpha y_A + \delta \lambda^2(\alpha+1))}{2}$ and $\Pi_B = \frac{y_B(-2\alpha y_B + \delta \lambda^2(\alpha+1))}{2}$. Since $\frac{\partial^2 \Pi_A}{\partial y_A^2} = \frac{\partial^2 \Pi_B}{\partial y_B^2} = -2\alpha < 0$, each profit function is concave in its own ad impression. The first-order conditions $\frac{\partial \Pi_A}{\partial y_A} = 0$ and $\frac{\partial \Pi_B}{\partial y_B} = 0$ yield $y_A^{CC} = y_B^{CC} = \frac{\delta \lambda^2(\alpha+1)}{4\alpha}$ and $\Pi_A^{CC} = \Pi_B^{CC} = \frac{\delta^2 \lambda^4(\alpha+1)^2}{16\alpha}$. Substituting these ad impressions into the demand and profit expressions shows that, except for $d_{At}^{CN} > 0$ and $d_{Bt}^{CN} > 0$ requiring $\alpha > \frac{1}{3}$, all other quantities are unconditionally positive. Analogous to Lemma 2, when $\alpha < \frac{1}{3}$ the profit functions reduce to $\Pi_A = \lambda^2 \delta y_A \frac{(1-\alpha)}{2}$ and $\Pi_B = \lambda^2 \delta y_B \frac{(1-\alpha)}{2}$. Both publishers set $y_A^{CC} = y_B^{CC} = \lambda^2 \delta$ to maximize the capture of consumer surplus from AI users clicking on their respective links via the AI engine. In equilibrium, profits are $\Pi_A^{CC} = \Pi_B^{CC} = \frac{\lambda^4 \delta^2 (1-\alpha)}{2}$, provided that $\alpha < \frac{1}{3}$.

Proof of Proposition 1.

By comparing publishers' profits across subgames over $\alpha \in (0, \frac{1}{3}]$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, and $\alpha \in (\frac{1}{2}, 1)$, we characterize equilibrium copyright openness. Publishers are symmetric, so CN is equivalent to NC. From Lemma 2, equilibrium profits in the NC case satisfy $\Pi_A^{NC} = \frac{\lambda^2 \delta^2 \alpha}{4}$ and $\Pi_B^{NC} = \frac{\delta^2 \lambda^2}{4\alpha}$ for $\alpha > \frac{1}{2}$, while $\Pi_A^{NC} = \frac{\lambda^2 \delta^2 \alpha}{4}$ and $\Pi_B^{NC} = \lambda^2 \delta^2 (1 - \alpha)$ for $\alpha < \frac{1}{2}$.

(1) When $\alpha \in (\frac{1}{2}, 1)$: NN is an equilibrium if $\Pi_A^{NN} - \Pi_A^{CN} \geq 0$ and $\Pi_B^{NN} - \Pi_B^{NC} \geq 0$. Since $\Pi_A^{NN} - \Pi_A^{CN} = \Pi_B^{NN} - \Pi_B^{NC} = \frac{\alpha}{2} - \frac{\delta^2 \lambda^2}{4\alpha} > 0$ if and only if $\delta < \dot{\delta}_1 = \frac{\sqrt{2}\alpha}{\lambda}$, NN is the equilibrium when $\delta < \dot{\delta}_1$. CN is an equilibrium if $\Pi_A^{CN} - \Pi_A^{NN} > 0$ and $\Pi_B^{CN} - \Pi_B^{CC} > 0$. Given that $\Pi_A^{NN} - \Pi_A^{CN} > 0$ holds when $\delta < \dot{\delta}_1$, we can obtain that $\Pi_A^{CN} - \Pi_A^{NN} > 0$ holds when $\delta > \dot{\delta}_1$. For publisher B, $\Pi_B^{CN} - \Pi_B^{CC} = \frac{\delta^2(-(\alpha+1)^2 \lambda^4 + 4\lambda^2 \alpha^2)}{16\alpha} > 0$ only if $\alpha > \alpha_1 = \frac{\lambda}{2-\lambda}$, which is infeasible given $\alpha > \frac{1}{2}$ and $\delta > \dot{\delta}_1$. Hence, CN is never an equilibrium. CC is an equilibrium if $\Pi_A^{CC} - \Pi_A^{NC} > 0$ and $\Pi_B^{CC} - \Pi_B^{CN} > 0$. Since $\Pi_B^{CN} - \Pi_B^{CC} > 0$ for $\alpha > \alpha_1$, we have $\Pi_A^{CC} - \Pi_A^{NC} = \Pi_B^{CC} - \Pi_B^{CN} > 0$ when $\alpha < \alpha_1$. Thus, CC and NN coexist for $\delta < \dot{\delta}_1$ and $\alpha < \alpha_1$. Because $\Pi_A^{NN} = \Pi_B^{NN} > \Pi_A^{CC} = \Pi_B^{CC}$, NN is Pareto-dominant and is selected in this region. Let $\alpha_0 = \frac{\lambda\sqrt{2}}{2}$ denote the solution to $\dot{\delta}_1 = 1$. Since $\alpha_0 < \alpha_1$ and $\dot{\delta}_1$ increases in α , CC is selected only when $\delta > \dot{\delta}_1$.

(2) When $\alpha \in (\frac{1}{3}, \frac{1}{2}]$: By analogous method for solving the equilibrium solution, NN is an

equilibrium if $\delta < \ddot{\delta}_1 = \frac{\sqrt{2}\sqrt{(1-\alpha)\alpha}}{2(1-\alpha)\lambda}$. For CN to be an equilibrium, $\delta > \ddot{\delta}_1$ and $\alpha > \alpha_1$ must both hold, which never occurs for $\frac{1}{3} < \alpha < \frac{1}{2}$. CC becomes the equilibrium only when $\alpha < \alpha_1$. CC and NN coexist for $\delta < \ddot{\delta}_1$ and $\alpha < \alpha_1$, where CC is dominated by NN. Let $\alpha_2 = \frac{2\lambda^2}{2\lambda^2+1}$ denote the solution to $\ddot{\delta}_1 = 1$. Since $\alpha_2 < \alpha_1$ and $\ddot{\delta}_1$ increases in α , CC is selected only when $\delta > \ddot{\delta}_1$.

(3) When $\alpha \in (0, \frac{1}{3}]$: Similarly, NN is an equilibrium if $\delta < \ddot{\delta}_1$. The conditions $\delta > \ddot{\delta}_1$ and $\alpha > \alpha_2$ required for CN never hold simultaneously for $0 < \alpha < \frac{1}{3}$. CC becomes the equilibrium only when $\alpha < \alpha_2$. Therefore, CC and NN coexist for $\delta < \ddot{\delta}_1$ and $\alpha < \alpha_2$, where CC is dominated by NN. Since $\ddot{\delta}_1$ increases in α and α_2 solves $\ddot{\delta}_1 = 1$, CC is selected only when $\delta > \ddot{\delta}_1$.

Thus, Proposition 1 is established. The corresponding threshold δ_1 equals $\ddot{\delta}_1$ for $\alpha \leq \frac{1}{2}$ and $\dot{\delta}_1$ for $\alpha > \frac{1}{2}$.

Proof of Corollary 1.

We compare publishers' profits under the NN and CC cases for $\alpha \in (0, \frac{1}{3}]$ and $\alpha \in (\frac{1}{3}, 1)$. For $\alpha \in (\frac{1}{3}, 1)$, $\Pi_A^{NN} - \Pi_A^{CC} = \Pi_B^{NN} - \Pi_B^{CC} = \frac{\alpha}{2} - \frac{\delta^2\lambda^4(\alpha+1)^2}{16\alpha}$. Define the threshold $\dot{\delta}_2 = \frac{2\sqrt{2}\alpha}{(\alpha+1)\lambda^2}$. The difference is positive if $\delta < \dot{\delta}_2$ and negative otherwise. For $\alpha \in (0, \frac{1}{3}]$, $\Pi_A^{NN} - \Pi_A^{CC} = \Pi_B^{NN} - \Pi_B^{CC} = \frac{\alpha}{2} - \frac{\lambda^4\delta^2(1-\alpha)}{2}$. Define the threshold $\ddot{\delta}_2 = -\frac{\sqrt{-(\alpha-1)\alpha}}{(\alpha-1)\lambda^2}$. The difference is positive if $\delta < \ddot{\delta}_2$ and negative otherwise. Therefore, $\delta_2 = \dot{\delta}_2$ for $\alpha > \frac{1}{3}$ and $\delta_2 = \ddot{\delta}_2$ for $\alpha \leq \frac{1}{3}$, establishing Corollary 1.

Proof of Proposition 2.

We compare welfare outcomes under the NN and CC cases. Note that the consumer surplus (denoted by CS) in the NN case is $CS^{NN} = \alpha(\int_0^{x_{NN}} (v - y_A - x)dx + \int_{x_{NN}}^1 (v - y_B - (1-x))dx) + (1-\alpha)(v - \delta) = \frac{(4\alpha-4)\delta}{4} + v - \frac{5\alpha}{4}$. In the CC case, when $\alpha > \frac{1}{3}$, $CS^{CC} = \alpha(\int_0^{x_A^{CC}} (v - y_A - x)dx + \int_{x_A^{CC}}^{x_B^{CC}} (v - \lambda^2\delta)dx + \int_{x_B^{CC}}^1 (v - y_B - (1-x))dx) + (1-\alpha)(\lambda^2\delta(v - y_A/y_B) + (1-\lambda^2\delta)(v - \lambda^2\delta)) = \frac{-3\delta^2\lambda^4\alpha^2 + (10\delta^2\lambda^4 - 16\lambda^2\delta + 16v)\alpha - 3\delta^2\lambda^4}{16\alpha}$. When $\alpha \leq \frac{1}{3}$, $CS^{CC} = \alpha(\int_0^1 (v - \lambda^2\delta)dx) + (1-\alpha)(\lambda^2\delta(v - y_A/y_B) + (1-\lambda^2\delta)(v - \lambda^2\delta)) = v - \lambda^2\delta$. Therefore, comparing the consumer surplus between NN and CC for $\alpha > \frac{1}{3}$, we get $CS^{NN} - CS^{CC} = \frac{(3\delta^2\lambda^4 + 16\delta - 20)\alpha^2 + (-10\delta^2\lambda^4 + 16\lambda^2\delta - 16\delta)\alpha + 3\delta^2\lambda^4}{16\alpha} < 0$. For $\alpha \leq \frac{1}{3}$, we get $CS^{NN} - CS^{CC} = (\lambda^2 + \alpha - 1)\delta - \frac{5\alpha}{4} < 0$. Hence, consumer surplus is always higher in the CC case than in the NN case.

Next, we compare the social welfare (denoted by SW) in cases NN and CC. In the NN case, $SW^{NN} = CS^{NN} + \Pi_A^{NN} + \Pi_B^{NN} = \frac{(4\alpha-4)\delta}{4} + v - \frac{\alpha}{4}$. In the CC case, for $\alpha > \frac{1}{3}$, $SW^{CC} = CS^{CC} + \Pi_A^{CC} + \Pi_B^{CC} = \frac{-3\delta^2\lambda^4\alpha^2 + (10\delta^2\lambda^4 - 16\lambda^2\delta + 16v)\alpha - 3\delta^2\lambda^4}{16\alpha} + \frac{\delta^2\lambda^4(\alpha+1)^2}{16\alpha} + \frac{\delta^2\lambda^4(\alpha+1)^2}{16\alpha} = \frac{(14\delta^2\lambda^4 - 16\lambda^2\delta + 16v)\alpha - \delta^2\lambda^4(1+\alpha^2)}{16\alpha}$. For $\alpha \leq \frac{1}{3}$, $SW^{CC} = CS^{CC} + \Pi_A^{CC} + \Pi_B^{CC} = (v - \lambda^2\delta) + \frac{\lambda^4\delta^2(1-\alpha)}{2} + \frac{\lambda^4\delta^2(1-\alpha)}{2} = v - \lambda^4\delta^2(-1 + \alpha) - \lambda^2\delta$. Thus, comparing the difference for $\alpha \leq \frac{1}{3}$, we get $SW^{NN} - SW^{CC} = \lambda^4\delta^2(-1 + \alpha) + (\lambda^2 + \alpha - 1)\delta - \frac{\alpha}{4} < 0$. And for $\alpha > \frac{1}{3}$, we get $SW^{NN} - SW^{CC} = \frac{(\delta^2\lambda^4 + 16\delta - 4)\alpha^2 + (-14\delta^2\lambda^4 + 16\lambda^2\delta - 16\delta)\alpha + \delta^2\lambda^4}{16\alpha}$,

which is positive for $\delta_1^{\text{SW}} < \delta < \delta_2^{\text{SW}}$, and negative for $\delta > \delta_1^{\text{SW}}$ or $\delta < \delta_2^{\text{SW}}$. The thresholds δ_1^{SW} and δ_2^{SW} are given by $\delta_1^{\text{SW}} = \frac{2(-4\lambda^2 + \sqrt{(\alpha^2 - 14\alpha + 17)\lambda^4 + 32(\alpha - 1)\lambda^2 + 16(\alpha - 1)^2 - 4\alpha + 4})\alpha}{\lambda^4(\alpha^2 - 14\alpha + 1)}$ and $\delta_2^{\text{SW}} = \frac{2(4\lambda^2 + \sqrt{(\alpha^2 - 14\alpha + 17)\lambda^4 + 32(\alpha - 1)\lambda^2 + 16(\alpha - 1)^2 - 4\alpha + 4})\alpha}{\lambda^4(\alpha^2 - 14\alpha + 1)}$.

Proof of Proposition 3.

Similar to Lemmas 1–3, we also apply backward induction and first derive the second-stage equilibria for subgames NN, CN, and CC under paid AI. To ensure positive demand in the traditional market for both publishers, meaning that both loyal and migrating users exist, which can simultaneously focus on the region where publisher incentives and the social optimum are most misaligned under free AI, the indifferent point between loyal users and migrating users in the traditional market lies in $(0, 1)$. This requires the condition that $\max\{\frac{(r-2\lambda\delta)\alpha + \lambda\delta}{\alpha}, \frac{(r-3\lambda^2\delta)\alpha + \lambda^2\delta}{2\alpha}\} < s < \min\{\frac{(r-3\lambda\delta+2)\alpha + \lambda\delta}{2\alpha}, \frac{(r-3\lambda^2\delta+2)\alpha + \lambda^2\delta}{2\alpha}\}$ to be satisfied. For the NN case, the equilibrium ad impressions and profits coincide with Lemma 1 that are $\hat{y}_A^{\text{NN}} = \hat{y}_B^{\text{NN}} = 1$ and $\hat{\Pi}_A^{\text{NN}} = \hat{\Pi}_B^{\text{NN}} = \frac{\alpha}{2}$. For the CN case, the equilibrium ad impressions are $\hat{y}_A^{\text{CN}} = \frac{(r+s)\alpha + \delta\lambda}{2\alpha}$ and $\hat{y}_B^{\text{CN}} = \frac{\delta\lambda + s}{2}$, with equilibrium profits $\hat{\Pi}_A^{\text{CN}} = \frac{(r^2 + (-6\delta\lambda - 4s + 4)r + s^2)\alpha^2 + 2\delta\lambda(r+s)\alpha + \delta^2\lambda^2}{4\alpha}$ and $\hat{\Pi}_B^{\text{CN}} = \frac{(\delta\lambda + s)^2\alpha}{4}$. For the CC case, the equilibrium ad impressions and profits are $\hat{y}_A^{\text{CC}} = \hat{y}_B^{\text{CC}} = \frac{(\lambda^2\delta + r + 2s)\alpha + \lambda^2\delta}{4\alpha}$ and $\hat{\Pi}_A^{\text{CC}} = \hat{\Pi}_B^{\text{CC}} = \frac{(\delta^2\lambda^4 - 12\delta(r - \frac{s}{3})\lambda^2 + 3r^2 + (-8s + 8)r + 4s^2)\alpha^2 + 4\lambda^2\delta(\frac{\lambda^2\delta}{2} + r + s)\alpha + \delta^2\lambda^4}{16\alpha}$. Using these second-stage equilibria, we derive the first-stage equilibrium under paid AI by a method analogous to Proposition 1, yielding Proposition 3. To ensure the uniqueness and tractability of the equilibrium root in the publishers' copyright openness decisions, we also require $s < \frac{3r+8}{8}$. The thresholds are $\hat{\delta}_1 = \frac{(3\alpha r - r - s + \sqrt{9\alpha^2 r^2 - 6\alpha r^2 - 6\alpha r s + 6r s - 4r + 2})\alpha}{\lambda}$ and $\hat{\delta}_2 = \frac{(2((-3r+s)\alpha + r + s)\lambda - 4s\alpha + \sqrt{X_1})\alpha}{4\alpha^2\lambda - (\alpha + 1)^2\lambda^3}$, where $X_1 = (3(11\lambda^2 + 4)\alpha^2 - 30\alpha\lambda^2 + \lambda^2)r^2 - 16(((\lambda - 1)\alpha - \lambda)s + (\frac{\lambda}{2} + 1)\alpha + \frac{\lambda}{2})((\lambda - 2)\alpha + \lambda)r + 4s^2((\lambda - 2)\alpha + \lambda)^2$.

Next, we examine the conditions under which the prisoner's dilemma arises in the paid AI scenario by comparing the profits of publishers in the NN case with those in the CC case. For each publisher, the profit difference between not opening and opening its copyright to the paid AI company is that $\hat{\Pi}_A^{\text{NN}} - \hat{\Pi}_A^{\text{CC}} = \hat{\Pi}_B^{\text{NN}} - \hat{\Pi}_B^{\text{CC}} = \frac{\alpha}{2} - \frac{(\delta^2\lambda^4 - 12\delta(r - \frac{s}{3})\lambda^2 + 3r^2 + (-8s + 8)r + 4s^2)\alpha^2 + 4\lambda^2\delta(\frac{\lambda^2\delta}{2} + r + s)\alpha + \delta^2\lambda^4}{16\alpha} = \frac{(-\delta^2\lambda^4 + 12\delta(r - \frac{s}{3})\lambda^2 - 3r^2 + (8s - 8)r - 4s^2 + 8)\alpha^2 - 4\lambda^2\delta(\frac{\lambda^2\delta}{2} + r + s)\alpha - \delta^2\lambda^4}{16\alpha}$. Setting this difference equal to zero yields the threshold $\hat{\delta}_3 = \frac{((6r - 2s)\alpha + \sqrt{(33\alpha^2 - 30\alpha + 1)r^2 + ((-16s - 8)\alpha^2 - 16\alpha + 16s - 8)r + 8(\alpha + 1)^2 - 2r - 2s})\alpha}{\lambda^2(\alpha + 1)^2}$, such that $\hat{\Pi}_A^{\text{NN}} - \hat{\Pi}_A^{\text{CC}} = \hat{\Pi}_B^{\text{NN}} - \hat{\Pi}_B^{\text{CC}} > 0$ for $\delta < \hat{\delta}_3$. Thus, opening copyrights can make publishers worse off than when neither opens. By Proposition 3, both publishers open copyrights when $\hat{\delta}_1 < \delta < \hat{\delta}_2$. Hence, both publishers opening copyrights constitutes a prisoner's dilemma for $\hat{\delta}_1 < \delta < \min\{\hat{\delta}_2, \hat{\delta}_3\}$.

Proof of Proposition 4.

We compare consumer surplus under NN, CN, and CC in the paid AI scenario. In the NN case, consumer surplus is $\widehat{CS}^{NN} = \alpha(\int_0^{\hat{x}^{NN}} (v - y_A - x)dx + \int_{\hat{x}^{NN}}^1 (v - y_B - (1 - x))dx) + (1 - \alpha)(v - \delta - s) = \frac{(4\delta+4s-5)\alpha}{4} + v - s - \delta$, where $\hat{x}^{NN} = \frac{y_B}{2} + \frac{1}{2} - \frac{y_A}{2}$. In the CN case, $\widehat{CS}^{CN} = \alpha(\int_0^{\hat{x}_A^{CN}} (v - y_A - x)dx + \int_{\hat{x}_A^{CN}}^{\hat{x}_B^{CN}} (v - \lambda\delta - s)dx + \int_{\hat{x}_B^{CN}}^1 (v - y_B - (1 - x))dx) + (\lambda\delta(v - y_A) + (-\lambda\delta + 1)(v - \lambda\delta - s))(1 - \alpha) = \frac{(-3\delta^2\lambda^2+2\delta\lambda s+r^2-2rs+2s^2)\alpha^2+(8\delta^2\lambda^2-2\delta(r-s+4)\lambda-8s+8v)\alpha-3\delta^2\lambda^2}{8\alpha}$, where $\hat{x}_A^{CN} = \delta\lambda + s - y_A$ and $\hat{x}_B^{CN} = y_B - \delta\lambda - s + 1$. In the CC case, $\widehat{CS}^{CC} = \alpha(\int_0^{\hat{x}_A^{CC}} (v - y_A - x)dx + \int_{\hat{x}_A^{CC}}^{\hat{x}_B^{CC}} (v - \lambda^2\delta - s)dx + \int_{\hat{x}_B^{CC}}^1 (v - y_B - (1 - x))dx) + (\lambda^2\delta(v - y_A) + (1 - \lambda^2\delta)(v - \lambda^2\delta - s))(1 - \alpha) = \frac{(\delta\lambda^2+r-2s)(-3\delta\lambda^2+r-2s)\alpha^2+(10\delta^2\lambda^4-2\delta(r-2s+8)\lambda^2-16s+16v)\alpha-3\delta^2\lambda^4}{16\alpha}$, where $\hat{x}_A^{CC} = \lambda^2\delta + s - y_A$ and $\hat{x}_B^{CC} = y_B - \lambda^2\delta - s + 1$.

We first compare consumer surplus between case NN and CN and get the difference $\widehat{CS}^{NN} - \widehat{CS}^{CN} = \frac{(3\delta^2\lambda^2+(-2\lambda s+8)\delta-2s^2+(2r+8)s-r^2-10)\alpha^2+2(-4\delta\lambda^2-4+(r-s+4)\lambda)\delta\alpha+3\delta^2\lambda^2}{8\alpha}$. Setting this difference equal to zero yields the threshold $\hat{\delta}_0^{CS} = \frac{(-\sqrt{X_2}+(-s\alpha+r-s+4)\lambda+4\alpha-4)\alpha}{-(3\alpha^2-8\alpha+3)\lambda^2}$ such that $\widehat{CS}^{NN} > \widehat{CS}^{CN}$ for $\delta > \hat{\delta}_0^{CS}$ and the reverse otherwise. Next, comparing CC and CN gives $\widehat{CS}^{CC} - \widehat{CS}^{CN} = \frac{-3(\alpha-3)(\alpha-\frac{1}{3})\delta^2\lambda^4-2((r-2s-3\delta)\alpha^2+(r-2s+8\delta+8)\alpha-3\delta)\delta\lambda^2+4\alpha\delta(-s\alpha+r-s+4)\lambda-\alpha^2r^2}{16\alpha}$. Setting this equal to zero yields the threshold $\hat{\delta}_1^{CS} = \frac{\alpha(\sqrt{X_3}+(\alpha(r-2s)+r-2s+8)\lambda+2s\alpha-2r+2s-8)}{-3((\alpha^2-\frac{10}{3}\alpha+1)\lambda^2-2\alpha^2+\frac{16\alpha}{3}-2)\lambda}$, such that $\widehat{CS}^{CC} > \widehat{CS}^{CN}$ for $\delta > \hat{\delta}_1^{CS}$ and the reverse otherwise. Finally, comparing NN and CC yields $\widehat{CS}^{NN} - \widehat{CS}^{CC} = \frac{(-20+3\delta^2\lambda^4+2(8+(r-2s)\lambda^2)\delta-4s^2+4(r+4)s-r^2)\alpha^2+2\delta(-5\lambda^4\delta-8+(r-2s+8)\lambda^2)\alpha+3\delta^2\lambda^4}{16\alpha}$. Setting this equal to zero yields the threshold $\hat{\delta}_2^{CS} = \frac{(-2\sqrt{X_4}+(\alpha(r-2s)+r-2s+8)\lambda^2+8\alpha-8)\alpha}{-\lambda^4(3\alpha^2-10\alpha+3)}$, such that $\widehat{CS}^{NN} > \widehat{CS}^{CC}$ for $\delta > \hat{\delta}_2^{CS}$ and the reverse otherwise. Since $0 < \hat{\delta}_1^{CS} < \hat{\delta}_0^{CS} < \hat{\delta}_2^{CS} < 1$, Proposition 4 is established. The parameters in the above thresholds are defined as follows:

$$\begin{aligned} X_2 &= ((7s^2 - (6r + 24)s + 3r^2 + 30)\alpha^2 + ((14r + 56)s - 14s^2 - 8r^2 - 80)\alpha + 7s^2 - (8r + 32)s + 4r^2 + 8r + 46)\lambda^2 \\ &\quad + 8(-1 + \alpha)(-s\alpha + r - s + 4)\lambda + 16(-1 + \alpha)^2, \\ X_3 &= ((-2\lambda^2 + 6)\alpha^2 + (12\lambda^2 - 4\lambda - 16)\alpha - 2\lambda^2 - 4\lambda + 10)r^2 - 4(\lambda - 1)(\alpha\lambda + \lambda - 2)(s\alpha + s - 4)r \\ &\quad + 4(\lambda - 1)^2(s\alpha + s - 4)^2, \\ X_4 &= (((4s - 4r - 12)s + r^2 + 15)\alpha^2 + ((8r - 8s + 32)s - 2r^2 + 4r - 50)\alpha + 4s(s - r - 5) + r^2 + 4r + 31)\lambda^4 \\ &\quad + ((4r - 8s)\alpha^2 + 32\alpha - 4r + 8s - 32)\lambda^2 + 16(-1 + \alpha)^2. \end{aligned}$$

Proof of Proposition 5.

We compare social welfare (SW) across NN, CN, and CC under the paid AI scenario. In NN, $\widehat{SW}^{NN} = \widehat{CS}^{NN} + \widehat{\Pi}_A^{NN} + \widehat{\Pi}_B^{NN} = \frac{(4\delta+4s-1)\alpha}{4} - s + v - \delta$. In CN, $\widehat{SW}^{CN} = \widehat{CS}^{CN} + \widehat{\Pi}_A^{CN} + \widehat{\Pi}_B^{CN} = \frac{(-\delta^2\lambda^2-12(r-\frac{s}{2})\delta\lambda+3r^2+(-10s+8)r+6s^2)\alpha^2+(8\delta^2\lambda^2+2\delta(r+3s-4)\lambda-8s+8v)\alpha-\delta^2\lambda^2}{8\alpha}$. In CC, $\widehat{SW}^{CC} = \widehat{CS}^{CC} + \widehat{\Pi}_A^{CC} + \widehat{\Pi}_B^{CC} = \frac{(-\delta^2\lambda^4-26(r-\frac{6s}{13})\delta\lambda^2+7r^2+(-20s+16)r+12s^2)\alpha^2+(14\delta^2\lambda^4+6(r+2s-\frac{s}{3})\delta\lambda^2-16s+16v)\alpha-\delta^2\lambda^4}{16\alpha}$.

Then we conduct pairwise comparisons. For NN versus CC, we get the difference that $\widehat{SW}^{NN} - \widehat{SW}^{CC} = \frac{(\delta^2 \lambda^4 + (16 + (26r - 12s)\lambda^2)\delta - 7(r - 2s + 2)(r - \frac{6s}{7} + \frac{2}{7}))\alpha^2 - 6\delta(\frac{7\delta\lambda^4}{3} + \frac{8}{3} + (r + 2s - \frac{8}{3})\lambda^2)\alpha + \delta^2 \lambda^4}{16\alpha}$. Setting this difference equal to zero yields two thresholds that are $\hat{\delta}_1^{SW} = \frac{\alpha(8 + 6\lambda^2(\alpha + 1)s + \lambda^2(3 - 13\alpha)r - 8\lambda^2 + 2\sqrt{Y_1} - 8\alpha)}{\lambda^4(\alpha^2 - 14\alpha + 1)}$ and $\hat{\delta}_2^{SW} = \frac{\alpha(-8 - 6\lambda^2(\alpha + 1)s + \lambda^2(13\alpha - 3)r + 8\lambda^2 + 2\sqrt{Y_1} + 8\alpha)}{-\lambda^4(\alpha^2 - 14\alpha + 1)}$, such that $\widehat{SW}^{NN} > \widehat{SW}^{CC}$ for $\delta < \hat{\delta}_2^{SW}$ or $\delta > \hat{\delta}_1^{SW}$, and the reverse holds for $\hat{\delta}_2^{SW} < \delta < \hat{\delta}_1^{SW}$. For NN versus CN, $\widehat{SW}^{NN} - \widehat{SW}^{CN} = \frac{(\delta^2 \lambda^2 + (8 + (12r - 6s)\lambda)\delta - 3r^2 + (10s - 8)r - 6s^2 + 8s - 2)\alpha^2 - 2\delta(4\delta\lambda^2 + 4 + (r + 3s - 4)\lambda)\alpha + \delta^2 \lambda^2}{8\alpha}$. Setting this to zero yields $\hat{\delta}_0^{SW} = \frac{\alpha(3\lambda(\alpha + 1)s + (-6\alpha\lambda + \lambda)r + \sqrt{Y_2} - 4\alpha - 4\lambda + 4)}{(\alpha^2 - 8\alpha + 1)\lambda^2}$ and $\hat{\delta}_3^{SW} = \frac{\alpha(-3\lambda(\alpha + 1)s + (6\alpha\lambda - \lambda)r + \sqrt{Y_2} + 4\alpha + 4\lambda - 4)}{-(\alpha^2 - 8\alpha + 1)\lambda^2}$, such that $\widehat{SW}^{NN} > \widehat{SW}^{CN}$ for $\delta > \hat{\delta}_0^{SW}$ or $\delta < \hat{\delta}_3^{SW}$, and the reverse holds for $\hat{\delta}_3^{SW} < \delta < \hat{\delta}_0^{SW}$. For CC versus CN, $\widehat{SW}^{CC} - \widehat{SW}^{CN} = \frac{-\delta^2(\alpha^2 - 14\alpha + 1)\lambda^4 - 26\delta X_5 \lambda^2 + 24\delta\alpha((r - \frac{s}{2})\alpha - \frac{r}{6} - \frac{s}{2} + \frac{2}{3})\lambda + \alpha^2 r^2}{16\alpha}$, where $X_5 = (r - \frac{6s}{13} - \frac{\delta}{13})\alpha^2 + \frac{(8 - 3r - 6s + 8\delta)\alpha}{13} - \frac{\delta}{13}$. Setting this to zero yields the threshold $\hat{\delta}_4^{SW} = \frac{\alpha(\sqrt{2}\sqrt{85}\sqrt{Y_3} + ((-13r + 6s)\lambda + 12r - 6s)\alpha + (-8 + 3r + 6s)\lambda - 2r - 6s + 8)}{(\alpha^2 \lambda^2 - 14\alpha \lambda^2 - 2\alpha^2 + \lambda^2 + 16\alpha - 2)\lambda}$, such that $\widehat{SW}^{CC} > \widehat{SW}^{CN}$ for $\delta < \hat{\delta}_4^{SW}$ and $\widehat{SW}^{CC} < \widehat{SW}^{CN}$ for $\delta > \hat{\delta}_4^{SW}$. Finally, we compare the five thresholds above and therefore establish

Proposition 5. The parameters in these thresholds are defined as follows:

$$\begin{aligned} Y_1 &= (17 + (1 + 44r^2 + 4(1 - 11s)r + 12s^2 - 4s)\alpha^2 + 2(-7 - 22r^2 + 2(-1 + 10s)r - 12s^2 + 16s)\alpha + 4r^2 \\ &\quad + 4(-2 + s)r + 12s^2 - 28s)\lambda^4 + 4(-8 + (13r - 6s)\alpha^2 + 8(1 - 2r)\alpha + 3r + 6s)\lambda^2 + 16(\alpha - 1)^2, \\ Y_2 &= ((39r^2 + (-46s + 8)r + 15s^2 - 8s + 2)\alpha^2 + (-36r^2 + (50s - 16)r - 30s^2 + 40s - 16)\alpha + 4r^2 - 4rs + 15s^2 \\ &\quad - 32s + 18)\lambda^2 + 48\left(\left(r - \frac{s}{2}\right)\alpha - \frac{r}{6} - \frac{s}{2} + \frac{2}{3}\right)(\alpha - 1)\lambda + 16(\alpha - 1)^2, \\ Y_3 &= \frac{1}{85}((\lambda - 1)(85\lambda\alpha^2 r^2 - 78\lambda s\alpha^2 r + 18\lambda\alpha^2 s^2 - 71\alpha^2 r^2 + 72s\alpha^2 r - 18\alpha^2 s^2 - 46\lambda\alpha r^2 - 60\lambda\alpha r s + 36\lambda\alpha s^2 \\ &\quad + 16\alpha r^2 + 104\lambda\alpha r + 60\alpha r s - 48\lambda\alpha s - 36\alpha s^2 + 5\lambda r^2 + 18\lambda r s + 18\lambda s^2 - 96\alpha r + 48\alpha s - r^2 - 24\lambda r - 12rs \\ &\quad - 48\lambda s - 18s^2 + 16r + 32\lambda + 48s - 32)). \end{aligned}$$